

Model Analysis

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1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```
def build_df():
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()

    cols_to_check = target_col + features_X
    mask = np.isfinite(df[cols_to_check]).all(axis=1)
    df = df[mask]
```

```

cols = list(dict.fromkeys(target_col + features_X +
                          feats_to_balance + feats_grouping +
                          feats_reference+ feats_postproc))

df = df[cols]
#df['campaign'] = "ALL"

return df

```

```

def import_data_IR():
    if target_col[0] == "i_NH3":
        #file_path = "csv/innova/switched/df_signal_decomposition_NH3_current_db2.csv"
        file_path = "csv/innova/switched/df_signal_decomposition_current.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv/innova/switched/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)

    if target_col[0] == "i_NH3":
        df["campaign"] = df["datetime"].apply(assign_campaign)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/switched/features_decomposition_list.json", "r") as f:
            dict_features = json.load(f)
    elif target_col[0] == "i_CO2":
        df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/switched/features_decomposition_CO2_list.json", "r") as f:
            dict_features = json.load(f)
    else:
        raise ValueError("The target col do not match any existing function")

    df["hot"] = df.apply(assign_hot_season,axis=1)
    df["umidity_range"] = df.apply(assign_humidity, axis = 1)
    #Removing of rabbit 4

```

```
df["column"] = df.apply(assign_colonna, axis = 1)
df['height'] = df.apply(assign_altezza, axis = 1)
```

```
return df,dict_features
```

```
def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
    elif camp == "2025_Sep":
        mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
    elif camp == "2026_Gen":
        #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
        #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
        mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
    else:
        mapping = {}

    return mapping.get(rid, "no")
```

Function print :

1.2 Metadata

Nombre de lignes df	8116
Nombre de lignes df clean	8116

Nombre de features	29
% de NaN	0.0
Taille dataset entraînement	5681
Taille dataset test	2434
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_st_8h_mean, r_NH3_st_8h_std, r_NH3_st_8h_rms, r_NH3_st_8h_median, r_NH3_st_8h_min, r_NH3_st_8h_max, r_NH3_st_8h_diff_last, r_NH3_st_8h_slope, r_temperature_st_8h_mean, r_temperature_st_8h_std, r_temperature_st_8h_rms, r_temperature_st_8h_median, r_temperature_st_8h_min, r_temperature_st_8h_max, r_temperature_st_8h_diff_last, r_temperature_st_8h_slope, r_umidity_st_8h_mean, r_umidity_st_8h_std, r_umidity_st_8h_rms, r_umidity_st_8h_median, r_umidity_st_8h_min, r_umidity_st_8h_max, r_umidity_st_8h_diff_last, r_umidity_st_8h_slope

feats_to_balance :

feats_grouping :

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

2 Data correlation and clustering

2.1 Clustermat and cluster list

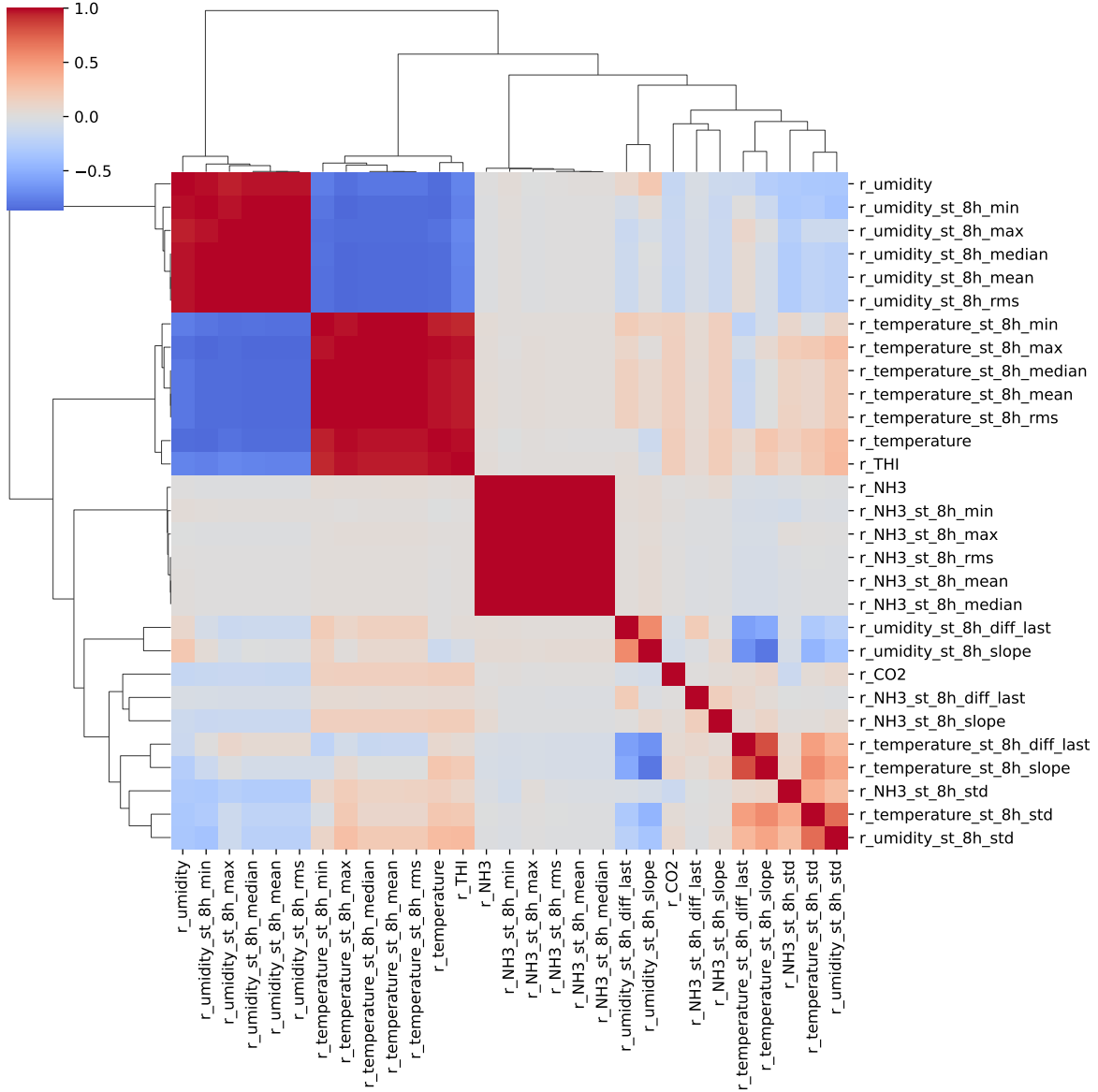
The following section has been done with a threshold choosen of : 0.8

cluster 1: r_NH3_st_8h_max, r_NH3_st_8h_rms, r_NH3, r_NH3_st_8h_min, r_NH3_st_8h_median, r_NH3_st_8h_mean ; **cluster 2:** r_umidity, r_umidity_st_8h_median, r_temperature_st_8h_rms, r_temperature_st_8h_max, r_temperature_st_8h_min, r_umidity_st_8h_rms, r_temperature_st_8h_median, r_THI, r_umidity_st_8h_mean, r_umidity_st_8h_max, r_umidity_st_8h_min, r_temperature, r_temperature_st_8h_mean ; **cluster 3:** r_CO2 ; **cluster 4:** r_NH3_st_8h_std ; **cluster 5:** r_NH3_st_8h_diff_last ; **cluster 6:** r_NH3_st_8h_slope ; **cluster 7:** r_temperature_st_8h_std ; **cluster 8:** r_temperature_st_8h_slope, r_umidity_st_8h_slope, r_temperature_st_8h_diff_last ; **cluster 9:** r_umidity_st_8h_std ; **cluster 10:** r_umidity_st_8h_diff_last ;

2.2 Features selection

Only the first items of the clusters is choosen for the rest of the study

Chooosen features_X: r_NH3_st_8h_max, r_umidity, r_CO2, r_NH3_st_8h_std, r_NH3_st_8h_diff_last, r_NH3_st_8h_slope, r_temperature_st_8h_std, r_temperature_st_8h_slope, r_umidity_st_8h_std, r_umidity_st_8h_diff_last



3 Model(s) analysis

3.1 Metadata

Features_X: r_NH3_st_8h_max, r_umidity, r_CO2, r_NH3_st_8h_std, r_NH3_st_8h_diff_last, r_NH3_st_8h_slope, r_temperature_st_8h_std, r_temperature_st_8h_slope, r_umidity_st_8h_std, r_umidity_st_8h_diff_last

Len(Features_X): 10

Config Gridsearch:{'show': True, 'search_type': 'optuna', 'model_name': None, 'model': None, 'param_grid': None, 'best_param': None, 'cv_func': ShuffleSplit(n_splits=5, random_state=42, test_size=0.3, train_size=None), 'save_models': True, 'sumup-table': True}

3.2 Explanation legend tables

Columns

- rank: ranking regarding cross validation test
- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv
- r_train: train_cv/test_cv

Colors:

[Orange] $r_{\text{test}} < 0.95$ (under performance of the test)

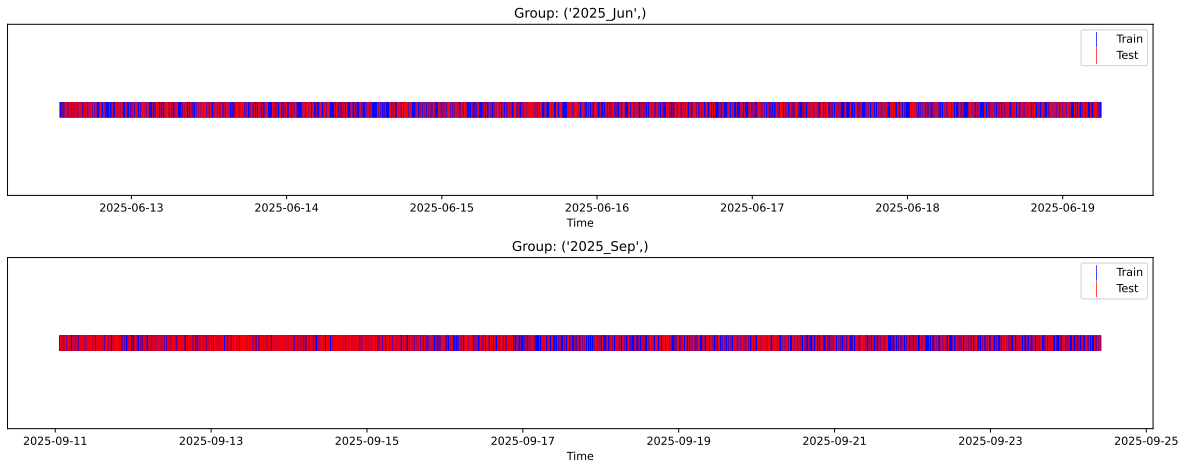
[Green] $r_{\text{test}} > 1.05$ (over performance of the test)

[Red] r_{train} not in $[0.95, 1.05]$ (gap between train/test)

3.3 Repartition of train and test

3.3.1 External Split: X_{train} vs X_{test}

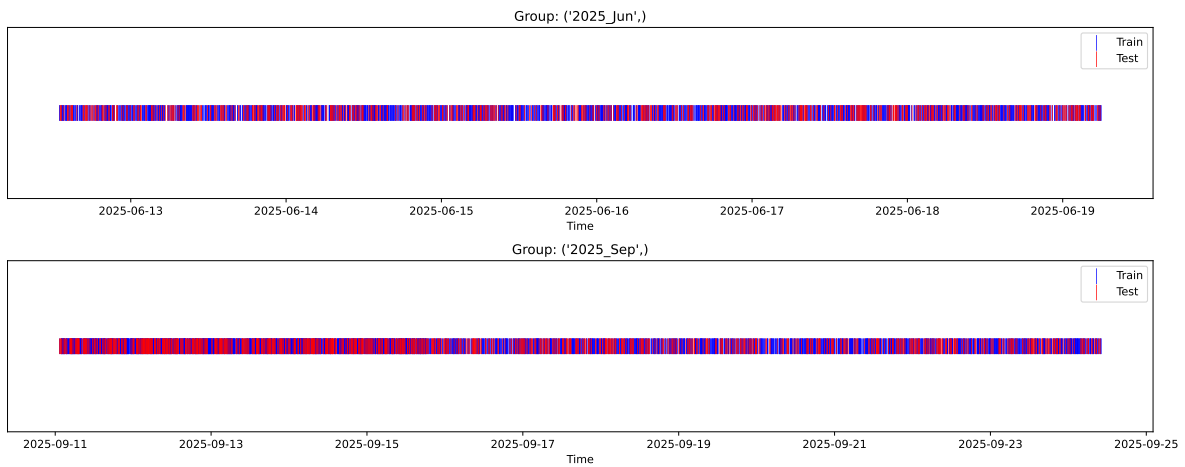
External Split: Train: 5681 samples Test: 2435 samples



3.3.2 Internal CV Folds

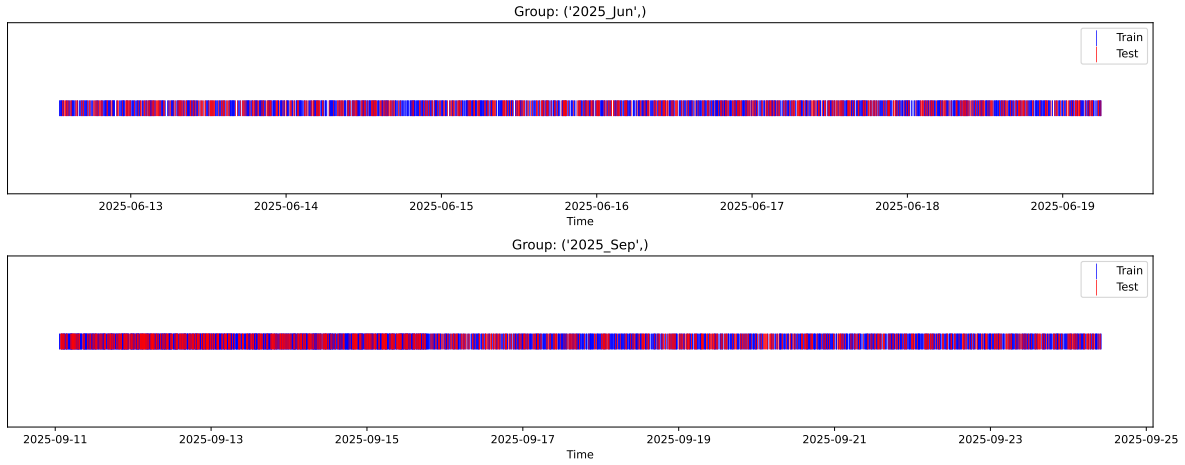
Fold 1/5

Train: 3976 samples Test: 1705 samples Excluded: 0 samples Window exclusion: 0h



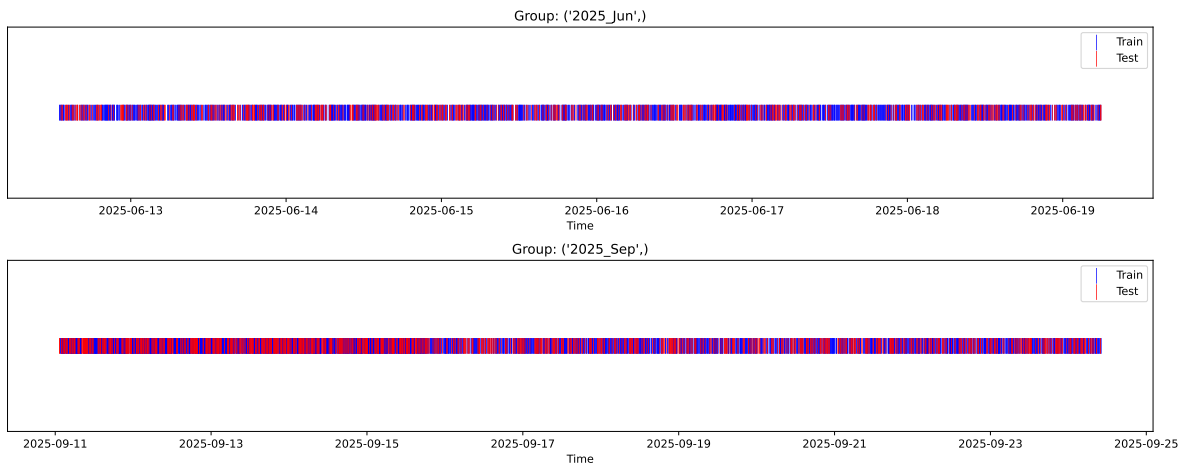
Fold 2/5

Train: 3976 samples Test: 1705 samples Excluded: 0 samples Window exclusion: 0h



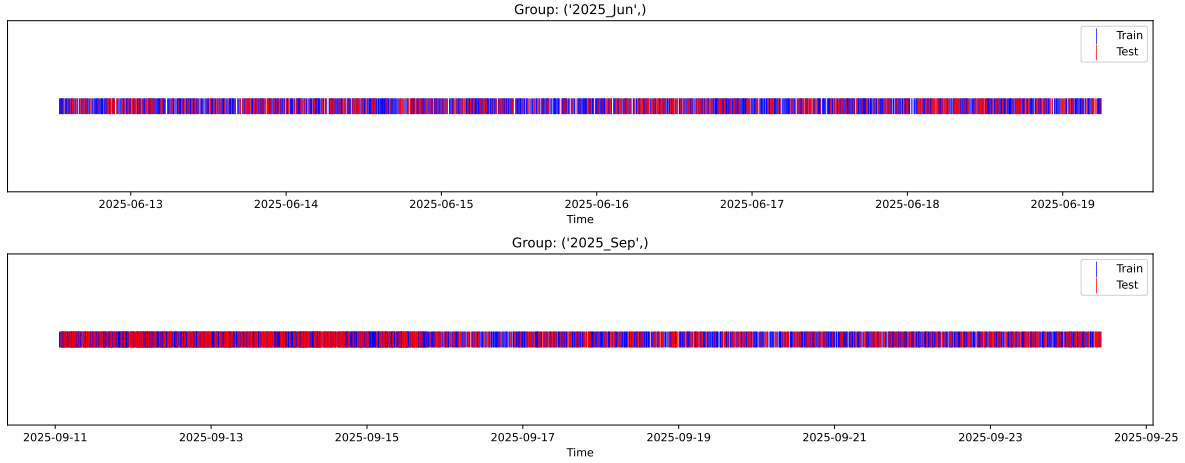
Fold 3/5

Train: 3976 samples Test: 1705 samples Excluded: 0 samples Window exclusion: 0h



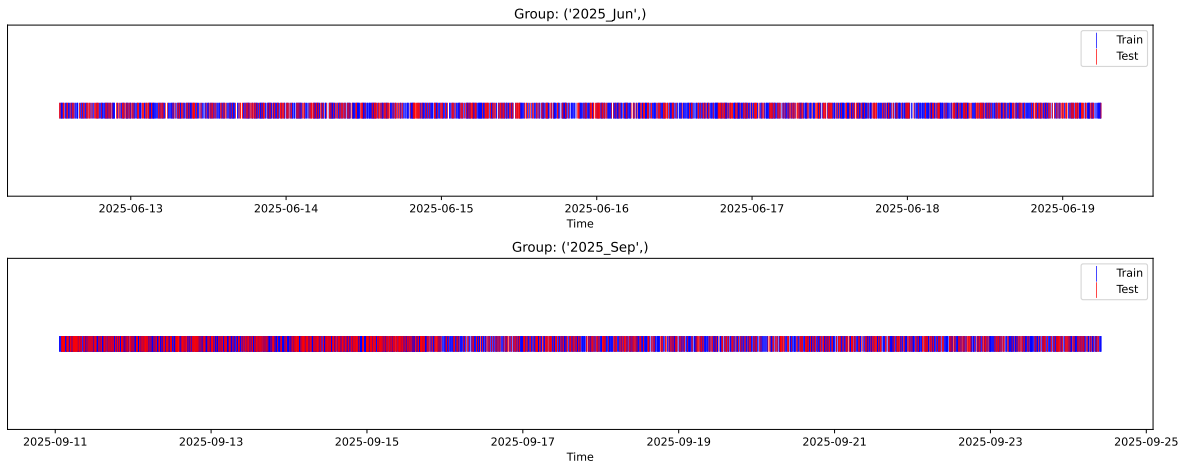
Fold 4/5

Train: 3976 samples Test: 1705 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 3976 samples Test: 1705 samples Excluded: 0 samples Window exclusion: 0h



3.4 Model : Linear

3.4.1 Gridsearch

Distribution y_{train} vs y_{test} : y_{train} : mean=2.345, std=0.790, min=0.959, max=8.682
 y_{test} : mean=2.364, std=0.810, min=1.041, max=13.845

Fold 0 — y_{test} : mean=2.32, std=0.75, y_{train} : mean=2.35, std=0.80

Fold 1 — y_{test} : mean=2.35, std=0.79, y_{train} : mean=2.34, std=0.79

Fold 2 — y_{test} : mean=2.34, std=0.80, y_{train} : mean=2.35, std=0.79

Fold 3 — y_test: mean=2.39, std=0.79, y_train: mean=2.32, std=0.79

Fold 4 — y_test: mean=2.36, std=0.81, y_train: mean=2.34, std=0.78

train scores CV: [0.52632813 0.5335071 0.5317195 0.54261781 0.53516441]

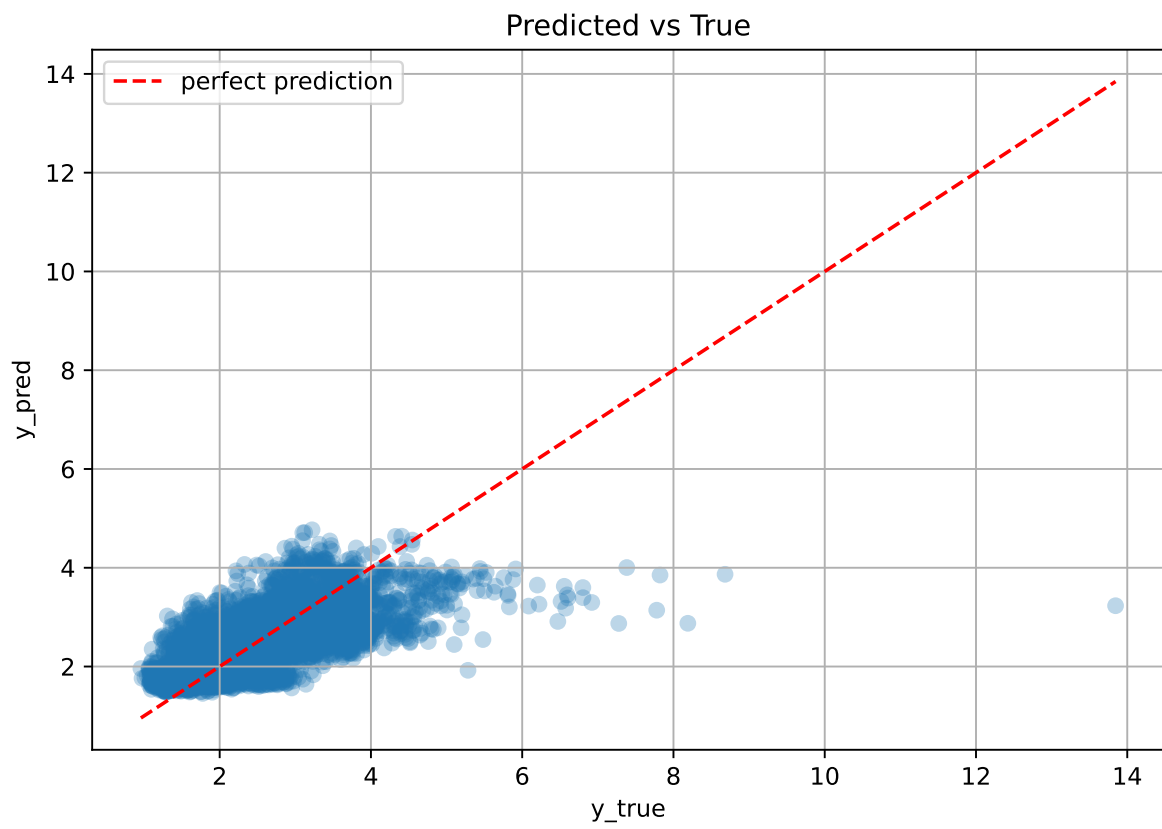
test scores CV: [0.54718854 0.53036095 0.53323726 0.50597741 0.52634898]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.5339	0.5286	0.0133	0.5031	0.5036	-0.025	0.9527	1.0099

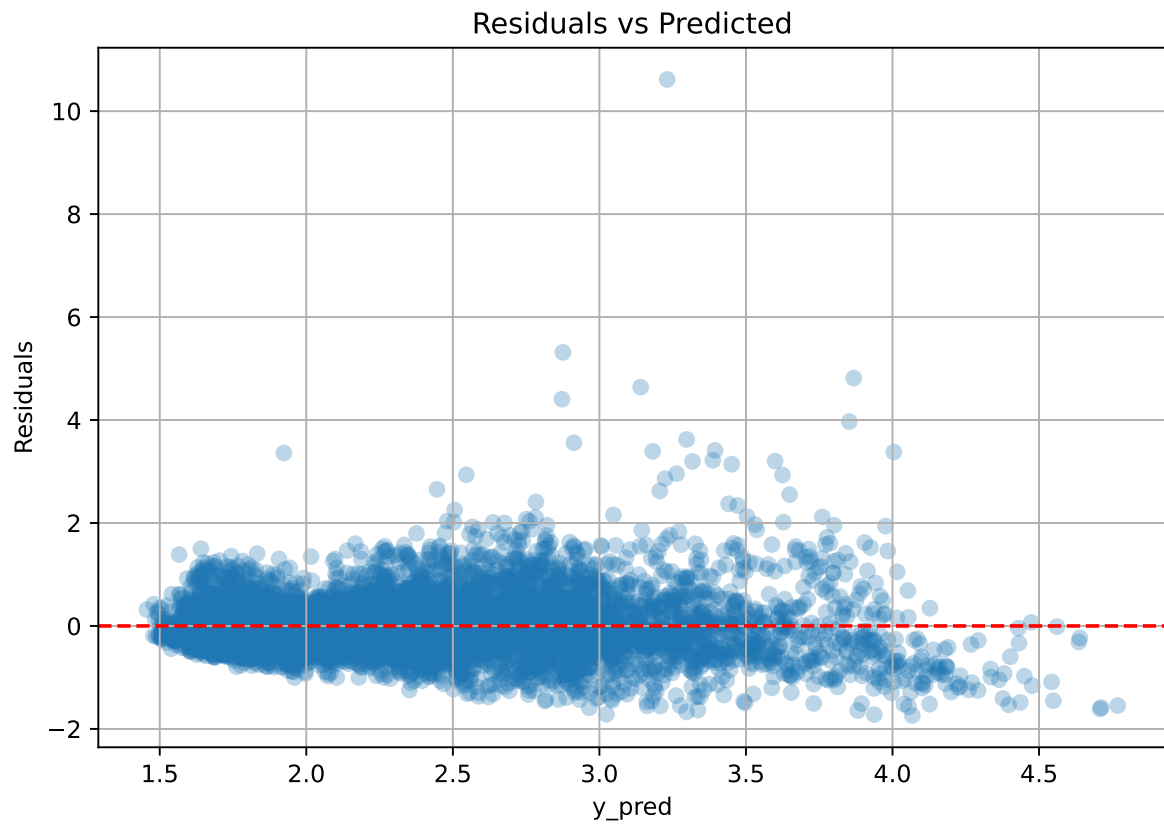
Coefficients:

coef_r_CO2 = -0.0426, coef_r_NH3_st_8h_diff_last = 0.0274, coef_r_NH3_st_8h_max = -0.0377, coef_r_NH3_st_8h_slope = 0.0140, coef_r_NH3_st_8h_std = 0.0526, coef_r_temperature_st_8h_slope = 0.1945, coef_r_temperature_st_8h_std = 0.0949, coef_r_umidity = -0.3470, coef_r_umidity_st_8h_diff_last = -0.0542, coef_r_umidity_st_8h_std = 0.0653, intercept = 2.3449,

3.4.2 Scatter



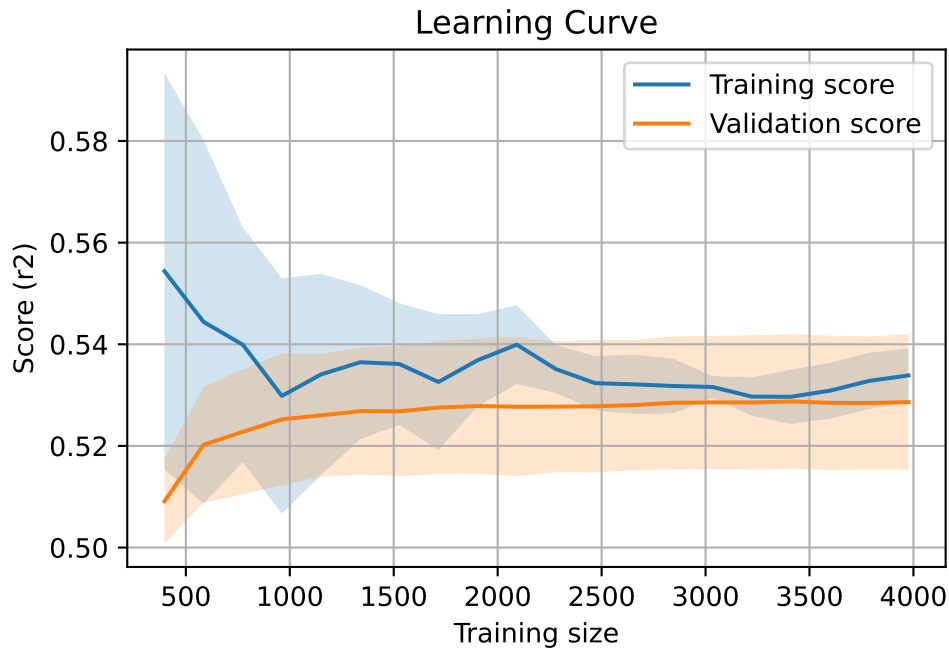
3.4.3 Scatter residuals



3.4.4 Learning curve

`Len(group)` : 8116, `Len(training)` : 5682,

`Len(training_real)` : 3978, `Len(test_of_training)` : 1704, `Len(test)` : 2434

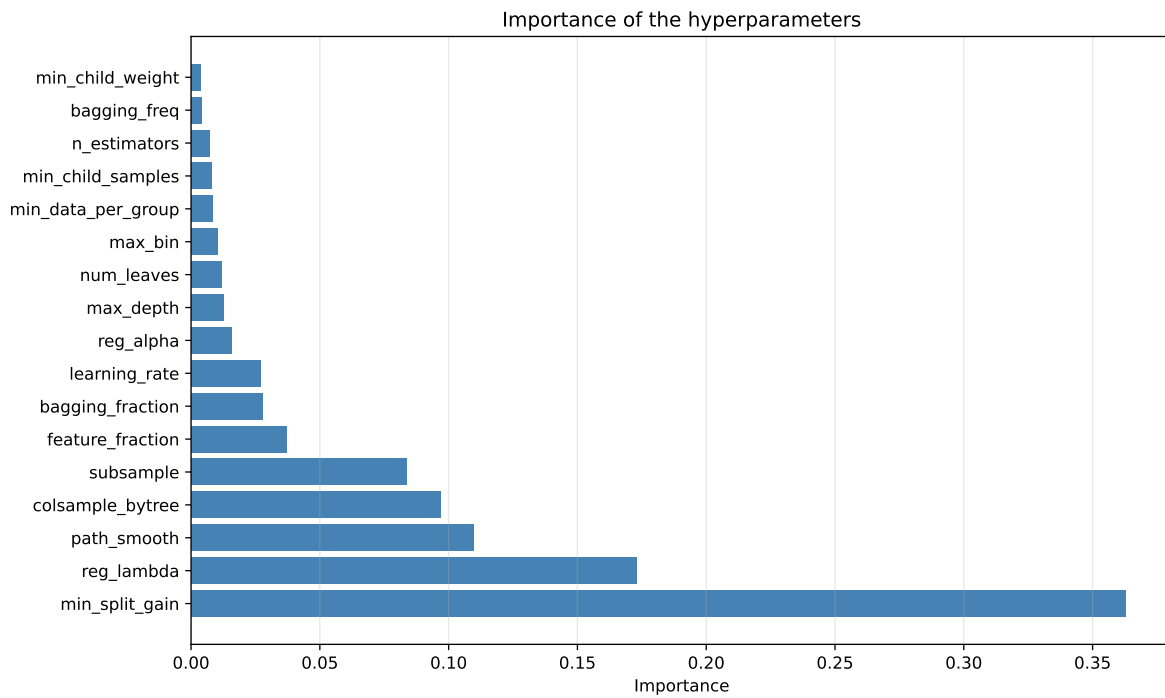
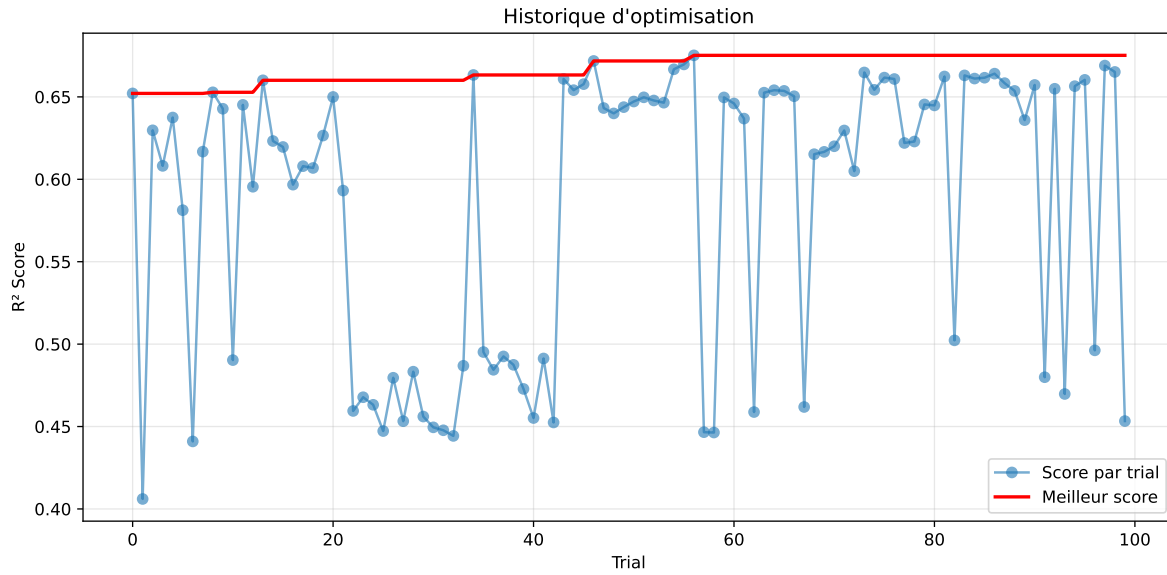


3.5 Model : LGBM

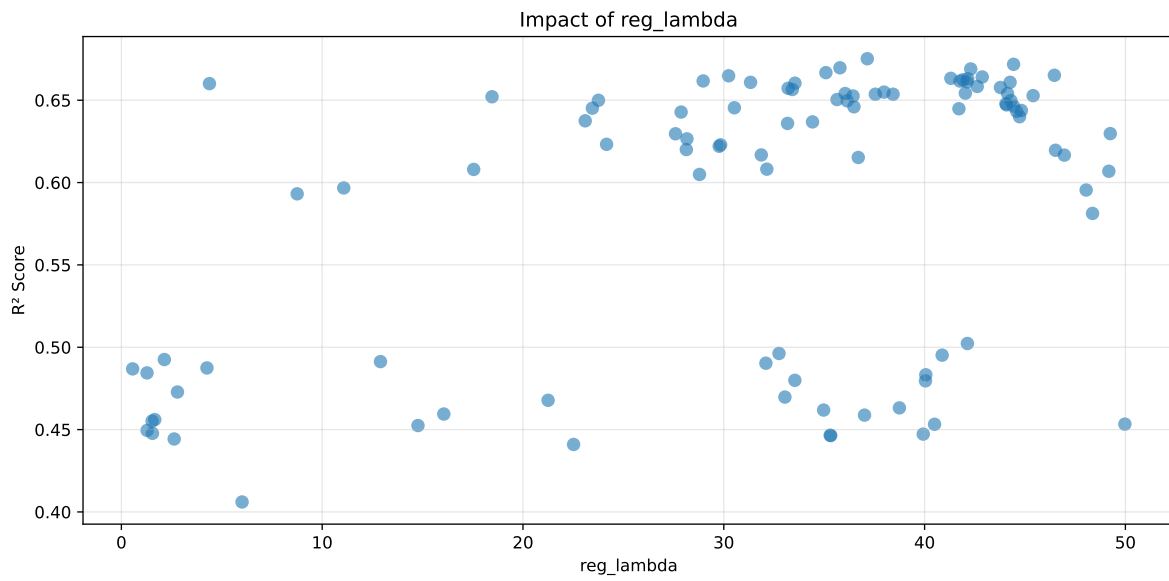
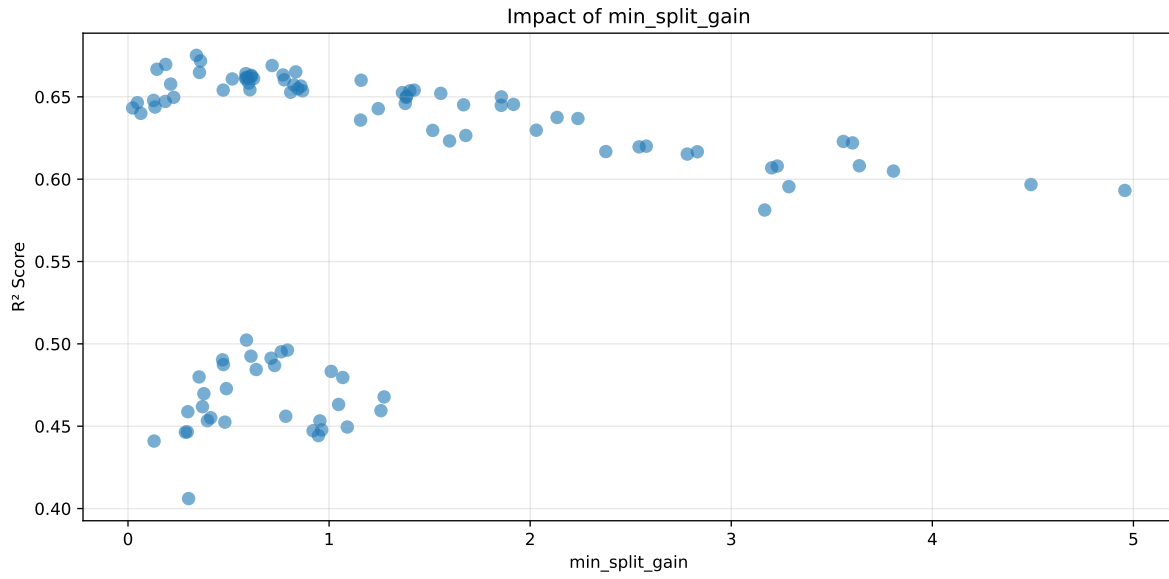
3.5.1 Gridsearch

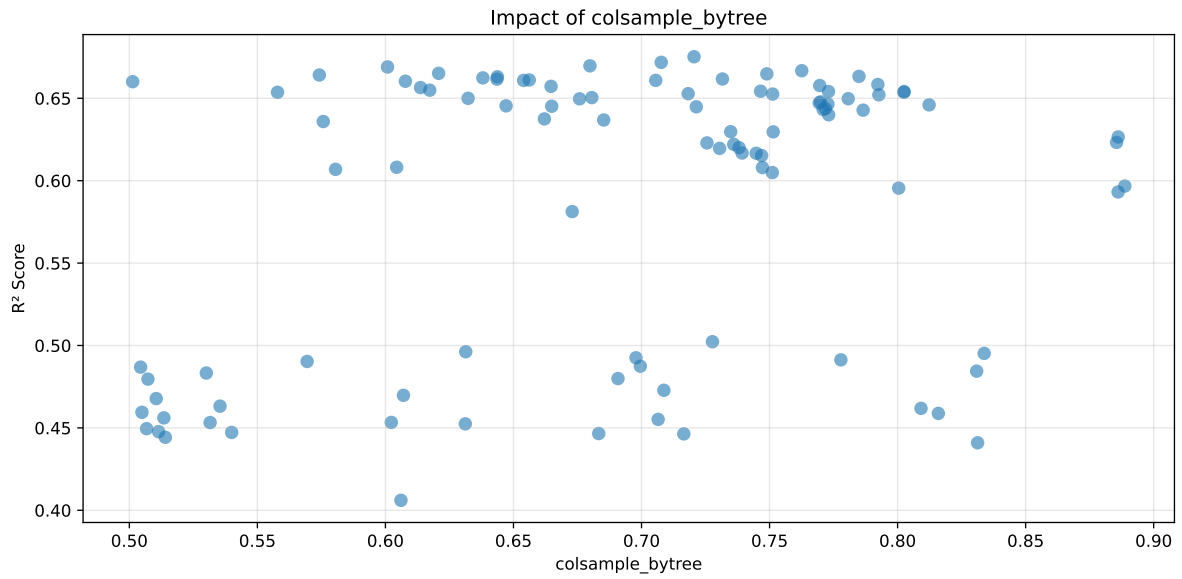
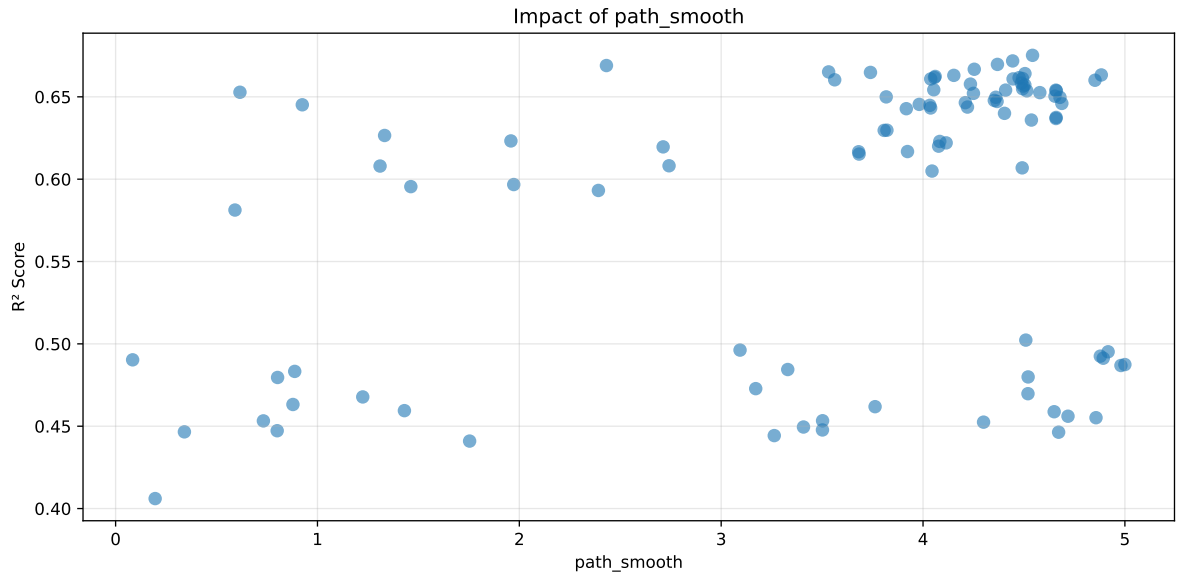
Distribution y_train vs y_test: y_train: mean=2.345, std=0.790, min=0.959, max=8.682
y_test: mean=2.364, std=0.810, min=1.041, max=13.845

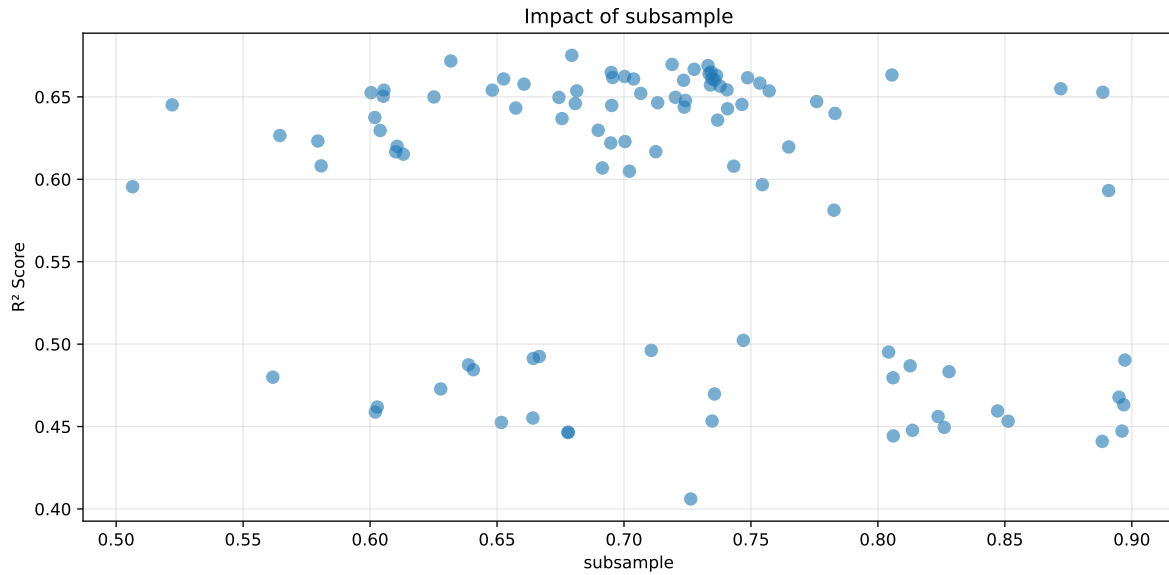
===== TOP Importance FEATURES =====
feature importance r_CO2 306 r_NH3_st_8h_max 277 r_umidity 239 r_tempera-
ture_st_8h_slope 162 r_NH3_st_8h_std 142 r_NH3_st_8h_slope 101 r_tempera-
ture_st_8h_std 91 r_NH3_st_8h_diff_last 88 r_umidity_st_8h_diff_last 82 r_umid-
ity_st_8h_std 70



Paramètres avec >5% d'importance : ['min_split_gain', 'reg_lambda', 'path_smooth', 'colsample_bytree', 'subsample']

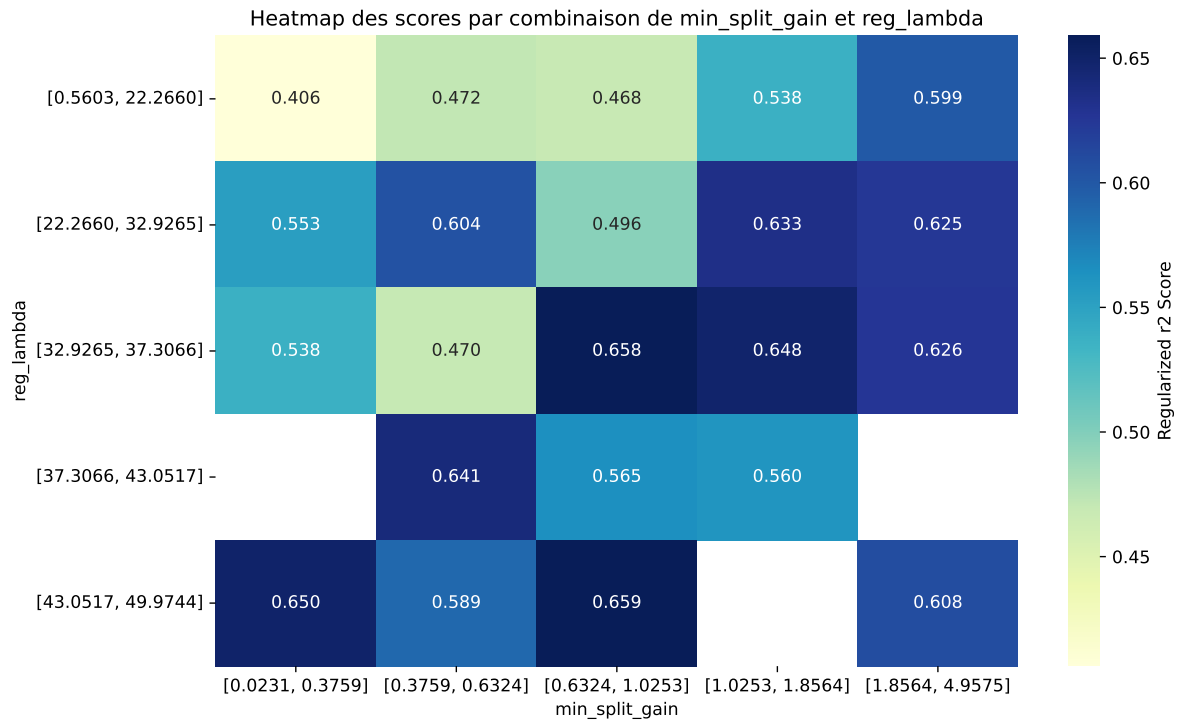






C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.7085	0.6752	0.6311	0.0118	0.6752	0.6463	-0.0289	0.9573	1.0493
2	0.7026	0.6718	0.6269	0.012	0.6718	0.637	-0.0348	0.9481	1.0458
3	0.7018	0.6697	0.6278	0.012	0.6697	0.637	-0.0327	0.9511	1.0479
4	0.7016	0.669	0.6251	0.0114	0.669	0.6411	-0.0279	0.9584	1.0487
5	0.6964	0.6667	0.6245	0.0116	0.6667	0.6333	-0.0335	0.9498	1.0445
6	0.6977	0.6651	0.6233	0.0114	0.6651	0.6401	-0.025	0.9623	1.0489
7	0.695	0.6648	0.6223	0.0121	0.6648	0.6337	-0.0312	0.9531	1.0454
8	0.6949	0.6641	0.6219	0.0112	0.6641	0.6351	-0.029	0.9564	1.0464
9	0.6934	0.6633	0.6207	0.012	0.6633	0.6414	-0.0219	0.967	1.0454
10	0.693	0.663	0.6224	0.0134	0.663	0.6379	-0.0251	0.9621	1.0452
---	---	---	---	---	---	---	---	---	---
100	0.7523	0.6946	0.6506	0.0085	0.4061	0.6668	-0.0278	0.96	1.0831

Hyperparameters:

Rank 1: {'reg__n_estimators': 800, 'reg__max_depth': 3, 'reg__learning_rate': 0.059127036186931124, 'reg__num_leaves': 36, 'reg__subsample': 0.6794548295618066, 'reg__colsample_bytree': 0.7205099754031388, 'reg__min_child_samples': 47, 'reg__reg_alpha': 2.613910020908771, 'reg__reg_lambda': 37.14694986141802, 'reg__min_split_gain': 0.34036550280609695, 'reg__path_smooth': 4.542502862741942, 'reg__min_child_weight': 0.7642177855557357, 'reg__feature_fraction': 0.788932218839396, 'reg__bagging_fraction': 0.7090146079766398, 'reg__bagging_freq': 3, 'reg__max_bin': 63, 'reg__min_data_per_group': 184}

Rank 2: {'reg__n_estimators': 700, 'reg__max_depth': 3, 'reg__learning_rate': 0.05245411066070164, 'reg__num_leaves': 35, 'reg__subsample': 0.631800307284297, 'reg__colsample_bytree': 0.7077034199428358, 'reg__min_child_samples': 44, 'reg__reg_alpha': 3.486563212567451, 'reg__reg_lambda': 44.431960067325136, 'reg__min_split_gain': 0.36076371813050945, 'reg__path_smooth': 4.4439542062074855, 'reg__min_child_weight': 0.525874678612557, 'reg__feature_fraction': 0.7308408671285557, 'reg__bagging_fraction': 0.6811749291228845, 'reg__bagging_freq': 2, 'reg__max_bin': 64, 'reg__min_data_per_group': 151}

Rank 3: {'reg__n_estimators': 700, 'reg__max_depth': 3, 'reg__learning_rate': 0.013477679239573048, 'reg__num_leaves': 36, 'reg__subsample': 0.7189344947551215, 'reg__colsample_bytree': 0.6798683283552311, 'reg__min_child_samples': 46, 'reg__reg_alpha': 2.510725915163522, 'reg__reg_lambda': 35.784703701840705, 'reg__min_split_gain': 0.18751958594863505, 'reg__path_smooth': 4.36867469429579, 'reg__min_child_weight': 0.8127965643181246, 'reg__feature_fraction': 0.7830718460955276, 'reg__bagging_fraction': 0.6914130579190889, 'reg__bagging_freq': 5, 'reg__max_bin': 65, 'reg__min_data_per_group': 184}

Rank 4: {'reg__n_estimators': 700, 'reg__max_depth': 4, 'reg__learning_rate': 0.04352946512569193, 'reg__num_leaves': 25, 'reg__subsample': 0.7330793993217227, 'reg__colsample_bytree': 0.6008157527285817, 'reg__min_child_samples': 47, 'reg__reg_alpha': 2.2262284053392007, 'reg__reg_lambda': 42.29829194190567, 'reg__min_split_gain': 0.7167648453425234, 'reg__path_smooth': 2.4316213341846655, 'reg__min_child_weight': 0.8783257687125106, 'reg__feature_fraction': 0.7484203837239388, 'reg__bagging_fraction': 0.7822496281549737, 'reg__bagging_freq': 7, 'reg__max_bin': 97, 'reg__min_data_per_group': 192}

Rank 5: {'reg__n_estimators': 700, 'reg__max_depth': 3, 'reg__learning_rate': 0.012347105180766314, 'reg__num_leaves': 35, 'reg__subsample': 0.7276447908744752, 'reg__colsample_bytree': 0.7625772751567139, 'reg__min_child_samples': 46, 'reg__reg_alpha': 3.898001270297078, 'reg__reg_lambda': 35.084842372118, 'reg__min_split_gain': 0.14390434063235968, 'reg__path_smooth': 4.253791584834646, 'reg__min_child_weight': 0.9067550221758174, 'reg__feature_fraction': 0.7916977609848931, 'reg__bagging_fraction': 0.6906814606967933, 'reg__bagging_freq': 1, 'reg__max_bin': 63, 'reg__min_data_per_group': 183}

Rank 6: {'reg__n_estimators': 700, 'reg__max_depth': 4, 'reg__learning_rate': 0.039117409191128744, 'reg__num_leaves': 32, 'reg__subsample': 0.7343151711070159, 'reg__colsample_bytree': 0.6207803139799296, 'reg__min_child_samples': 29, 'reg__reg_alpha': 1.870406755908578, 'reg__reg_lambda': 46.463413742264684, 'reg__min_split_gain': 0.8342462576847567, 'reg__path_smooth': 3.5323374740620754, 'reg__min_child_weight': 1.0782403393364266, 'reg__feature_fraction': 0.7701887738461081, 'reg__bagging_fraction': 0.7639199992062746, 'reg__bagging_freq': 7, 'reg__max_bin': 100, 'reg__min_data_per_group': 191}

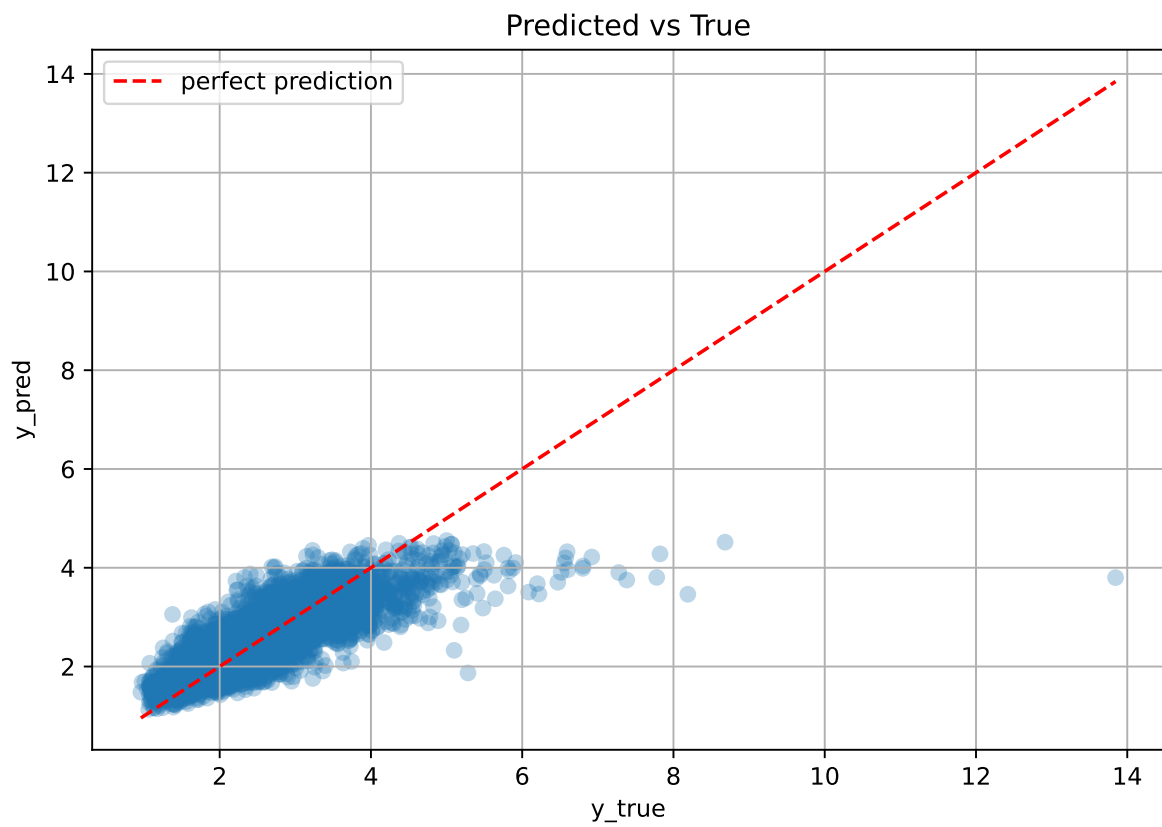
Rank 7: {'reg__n_estimators': 700, 'reg__max_depth': 3, 'reg__learning_rate': 0.013356893502845458, 'reg__num_leaves': 43, 'reg__subsample': 0.694974004636955, 'reg__colsample_bytree': 0.7489316261079212, 'reg__min_child_samples': 42, 'reg__reg_alpha': 4.840792934057627, 'reg__reg_lambda': 30.241198147210905, 'reg__min_split_gain': 0.35549520317460803, 'reg__path_smooth': 3.7396017078872337, 'reg__min_child_weight': 0.5490543757426881, 'reg__feature_fraction': 0.8227012329574561, 'reg__bagging_fraction': 0.6583694483593797, 'reg__bagging_freq': 3, 'reg__max_bin': 91, 'reg__min_data_per_group': 176}

Rank 8: {'reg__n_estimators': 700, 'reg__max_depth': 3, 'reg__learning_rate': 0.036721687248469664, 'reg__num_leaves': 38, 'reg__subsample': 0.7334878774961467, 'reg__colsample_bytree': 0.5741839334070167, 'reg__min_child_samples': 47, 'reg__reg_alpha': 2.070784784256202, 'reg__reg_lambda': 42.87024355223361, 'reg__min_split_gain': 0.5863239649562552, 'reg__path_smooth': 4.504969311467203, 'reg__min_child_weight': 0.004992516159273122, 'reg__feature_fraction': 0.7549336192086896, 'reg__bagging_fraction': 0.7479516343631529, 'reg__bagging_freq': 7, 'reg__max_bin': 98, 'reg__min_data_per_group': 190}

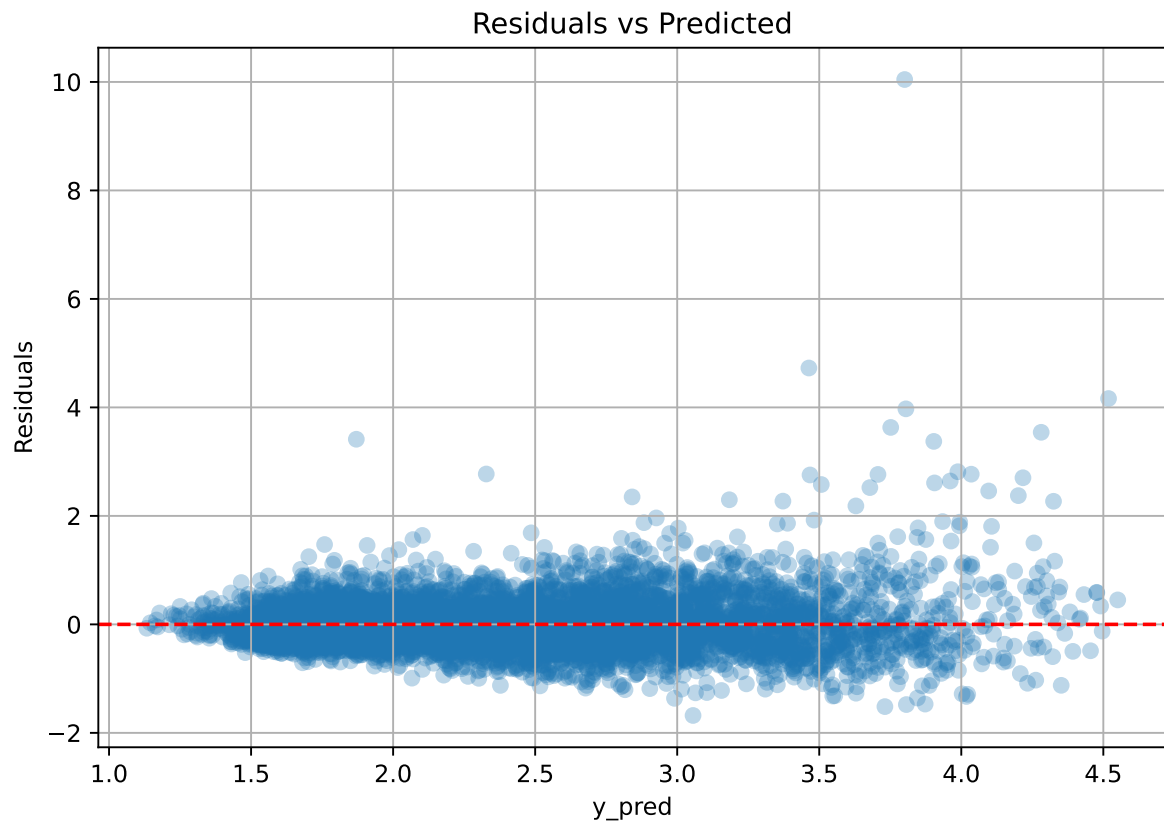
Rank 9: {'reg__n_estimators': 600, 'reg__max_depth': 4, 'reg__learning_rate': 0.026829231057183225, 'reg__num_leaves': 32, 'reg__subsample': 0.8055253837174898, 'reg__colsample_bytree': 0.7849186589390391, 'reg__min_child_samples': 49, 'reg__reg_alpha': 1.8621848818245375, 'reg__reg_lambda': 41.30862911394908, 'reg__min_split_gain': 0.771066539668634, 'reg__path_smooth': 4.883184467488414, 'reg__min_child_weight': 0.01657187532612032, 'reg__feature_fraction': 0.6634837476050874, 'reg__bagging_fraction': 0.6481754247761002, 'reg__bagging_freq': 4, 'reg__max_bin': 123, 'reg__min_data_per_group': 173}

Rank 10: {'reg__n_estimators': 700, 'reg__max_depth': 3, 'reg__learning_rate': 0.017177227872652863, 'reg__num_leaves': 29, 'reg__subsample': 0.7364475662855388, 'reg__colsample_bytree': 0.6437333723540751, 'reg__min_child_samples': 32, 'reg__reg_alpha': 2.0939222478415056, 'reg__reg_lambda': 42.15093570253765, 'reg__min_split_gain': 0.6129248295411432, 'reg__path_smooth': 4.152533710227354, 'reg__min_child_weight': 1.2733797856583107, 'reg__feature_fraction': 0.7570499713503405, 'reg__bagging_fraction': 0.7396778400325038, 'reg__bagging_freq': 7, 'reg__max_bin': 75, 'reg__min_data_per_group': 189}

3.5.2 Scatter



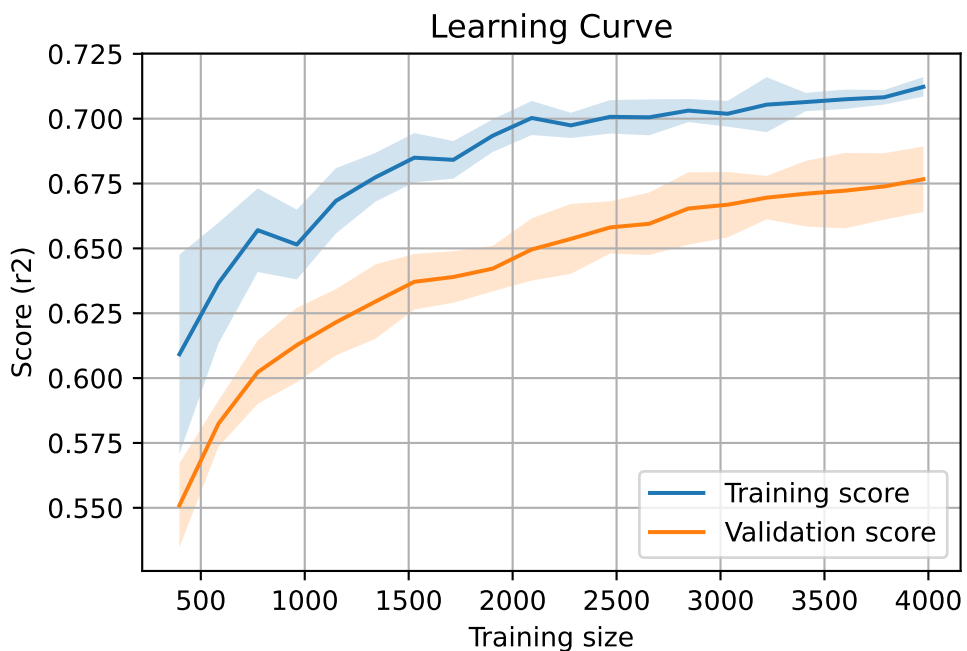
3.5.3 Scatter residuals



3.5.4 Learning curve

`Len(group)` : 8116, `Len(training)` : 5682,

`Len(training_real)` : 3978, `Len(test_of_training)` : 1704, `Len(test)` : 2434



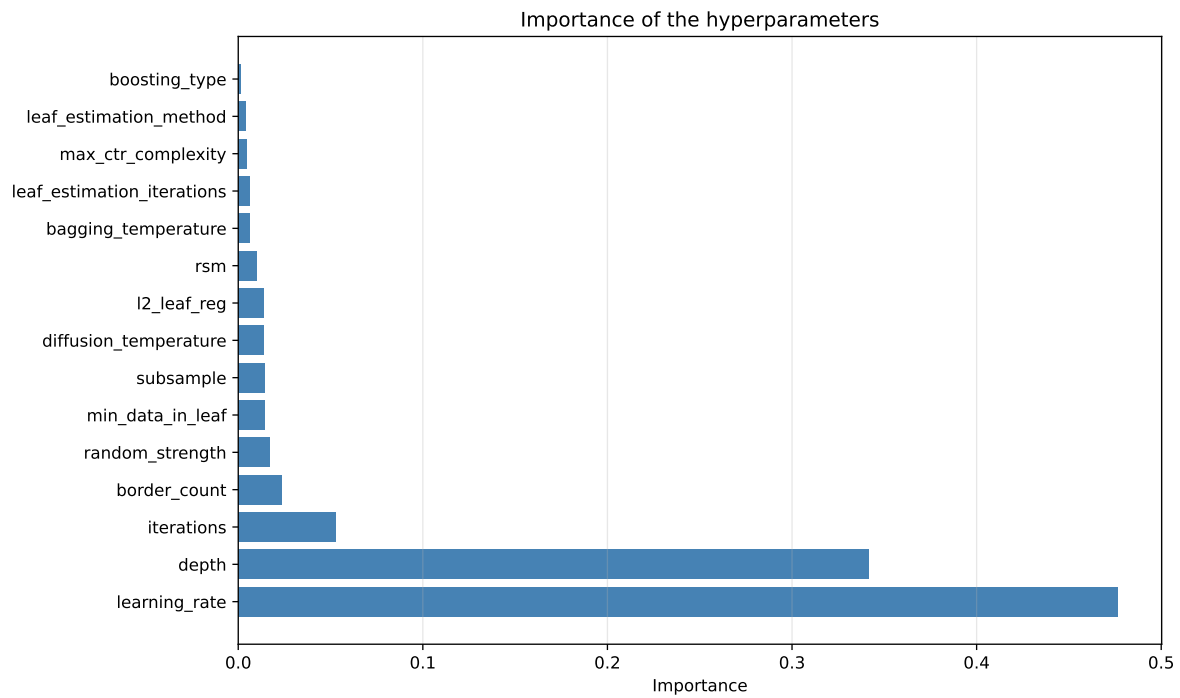
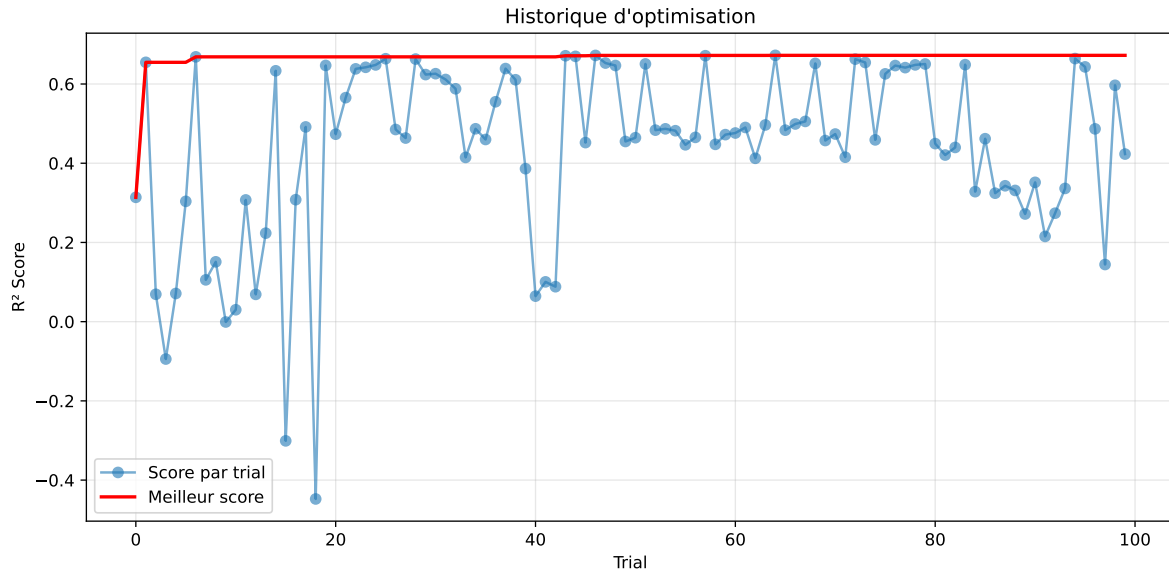
3.6 Model : CatBoost

3.6.1 Gridsearch

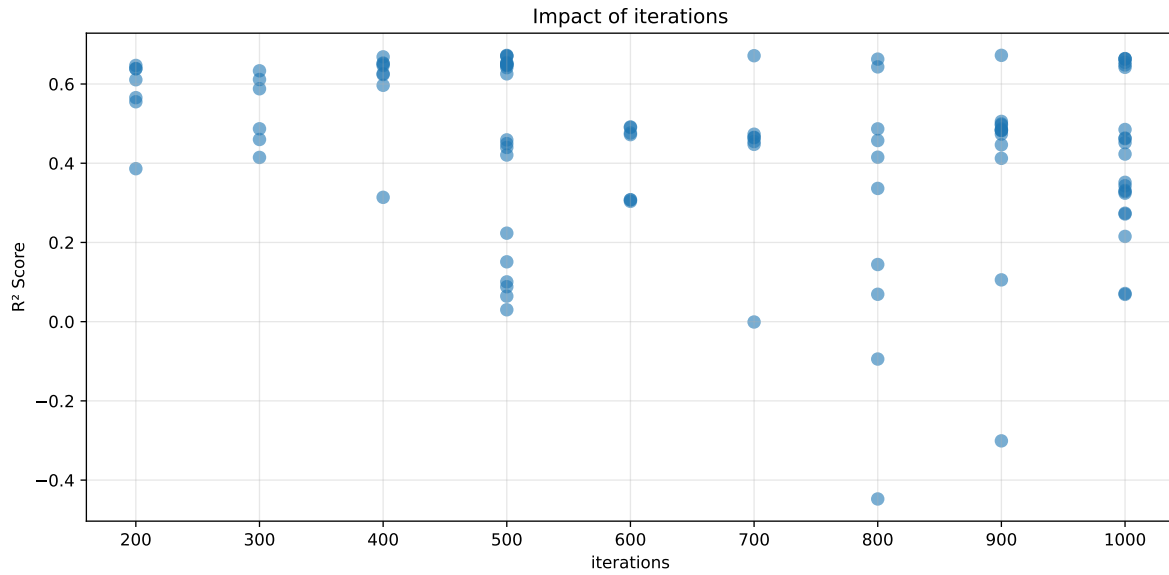
Distribution y_train vs y_test: y_train: mean=2.345, std=0.790, min=0.959, max=8.682
y_test: mean=2.364, std=0.810, min=1.041, max=13.845

===== TOP Importance FEATURES =====

feature	importance	r_umidity	34.895732	r_temperature_st_8h_slope	20.767389
r_CO2	12.145134	r_temperature_st_8h_std	9.387187	r_NH3_st_8h_max	8.133761
r_NH3_st_8h_std	3.925511	r_umidity_st_8h_std	3.423379	r_NH3_st_8h_slope	2.943048
r_umidity_st_8h_diff_last	2.729827	r_NH3_st_8h_diff_last	1.649032		

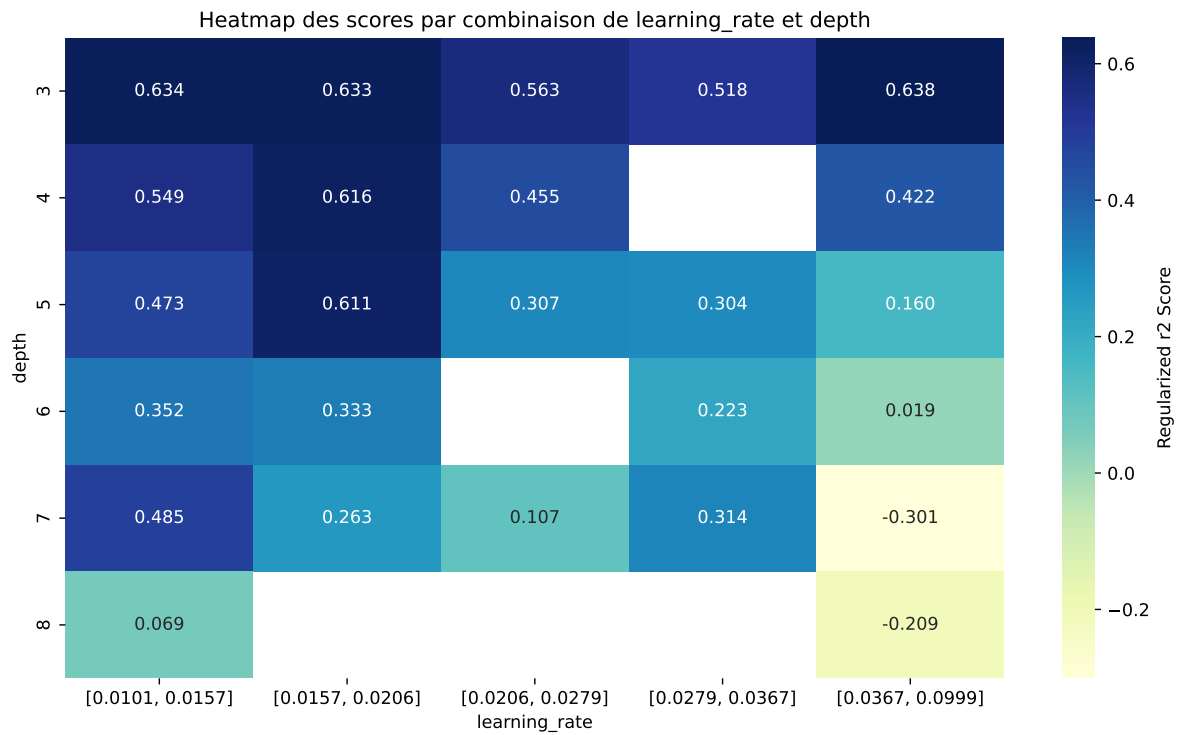


Paramètres avec >5% d'importance : ['learning_rate', 'depth', 'iterations']



C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.7036	0.6723	0.6308	0.0111	0.6723	0.6412	-0.0311	0.9537	1.0465
2	0.7004	0.6722	0.6273	0.0116	0.6722	0.6312	-0.041	0.939	1.042
3	0.7026	0.6715	0.6293	0.0104	0.6715	0.6323	-0.0392	0.9417	1.0464
4	0.701	0.6713	0.627	0.0108	0.6713	0.6314	-0.0399	0.9405	1.0442
5	0.6994	0.6698	0.6267	0.0098	0.6698	0.6317	-0.0381	0.9431	1.0441
6	0.6999	0.6686	0.6255	0.0109	0.6686	0.6296	-0.0389	0.9418	1.0468
7	0.6923	0.6642	0.6233	0.0115	0.6642	0.625	-0.0392	0.941	1.0423
8	0.6907	0.6638	0.6222	0.0098	0.6638	0.6281	-0.0357	0.9462	1.0405
9	0.6882	0.663	0.6221	0.0109	0.663	0.6295	-0.0335	0.9495	1.038
10	0.6885	0.6628	0.6215	0.0104	0.6628	0.6261	-0.0367	0.9447	1.0388
---	---	---	---	---	---	---	---	---	---
100	0.9916	0.7517	0.7088	0.0085	-0.4476	0.7244	-0.0274	0.9636	1.3191

Hyperparameters:

Rank 1: {'reg__iterations': 900, 'reg__depth': 3, 'reg__learning_rate': 0.022547634875267022, 'reg__l2_leaf_reg': 38.35330783297564, 'reg__bagging_temperature': 3.4590668712142834, 'reg__random_strength': 4.514738083733751, 'reg__subsample': 0.8348163728293481, 'reg__min_data_in_leaf': 48, 'reg__leaf_estimation_iterations': 4, 'reg__rsm': 0.6058459774602256, 'reg__border_count': 37, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 5450.952816288988}

Rank 2: {'reg__iterations': 500, 'reg__depth': 3, 'reg__learning_rate': 0.031012436545154023, 'reg__l2_leaf_reg': 27.255803008614304, 'reg__bagging_temperature': 2.3173763294160894, 'reg__random_strength': 3.7992260767215447, 'reg__subsample': 0.7389800181495219, 'reg__min_data_in_leaf': 50, 'reg__leaf_estimation_iterations': 9, 'reg__rsm': 0.7555765441684058, 'reg__border_count': 34, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 1825.6414412073054}

Rank 3: {'reg__iterations': 700, 'reg__depth': 3, 'reg__learning_rate': 0.02228880746560406, 'reg__l2_leaf_reg': 44.09629036919665, 'reg__bagging_temperature': 4.986282583328617, 'reg__random_strength': 3.112391577987445, 'reg__subsample': 0.6980452569453242, 'reg__min_data_in_leaf': 47, 'reg__leaf_estimation_iterations': 9, 'reg__rsm': 0.674655477861538, 'reg__border_count': 46, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 1199.238432718399}

Rank 4: {'reg__iterations': 500, 'reg__depth': 3, 'reg__learning_rate': 0.030863804328029397, 'reg__l2_leaf_reg': 24.278067615760907, 'reg__bagging_temperature': 4.988578437737461, 'reg__random_strength': 3.9597677515298826, 'reg__subsample': 0.7454120169850137, 'reg__min_data_in_leaf': 50, 'reg__leaf_estimation_iterations': 9, 'reg__rsm':

0.7701242691751832, 'reg__border_count': 35, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 3239.661586823564}

Rank 5: {'reg__iterations': 500, 'reg__depth': 3, 'reg__learning_rate': 0.03073800549589362, 'reg__l2_leaf_reg': 45.087508597163946, 'reg__bagging_temperature': 4.815621734782496, 'reg__random_strength': 3.902416177566697, 'reg__subsample': 0.7367667963210807, 'reg__min_data_in_leaf': 43, 'reg__leaf_estimation_iterations': 9, 'reg__rsm': 0.77307006992606, 'reg__border_count': 32, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 5013.021612810893}

Rank 6: {'reg__iterations': 400, 'reg__depth': 3, 'reg__learning_rate': 0.027464227515165834, 'reg__l2_leaf_reg': 39.288648017699806, 'reg__bagging_temperature': 4.044884004021636, 'reg__random_strength': 0.6117036889777511, 'reg__subsample': 0.6934988196638728, 'reg__min_data_in_leaf': 11, 'reg__leaf_estimation_iterations': 6, 'reg__rsm': 0.8796269444111424, 'reg__border_count': 99, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 9242.993477310974}

Rank 7: {'reg__iterations': 1000, 'reg__depth': 3, 'reg__learning_rate': 0.014007467540811919, 'reg__l2_leaf_reg': 32.04896835509439, 'reg__bagging_temperature': 3.6523874543830708, 'reg__random_strength': 4.103774768517557, 'reg__subsample': 0.7707220880148512, 'reg__min_data_in_leaf': 28, 'reg__leaf_estimation_iterations': 7, 'reg__rsm': 0.6332245728622938, 'reg__border_count': 128, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 8438.382154061012}

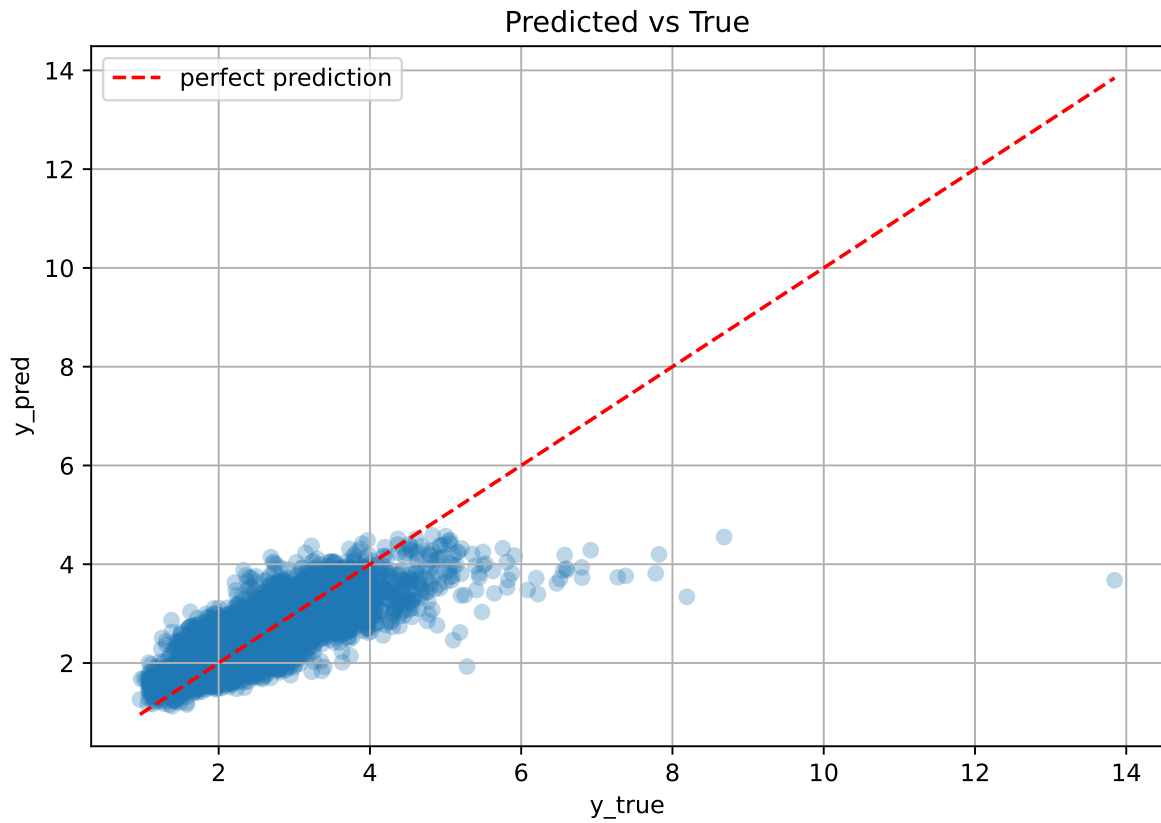
Rank 8: {'reg__iterations': 1000, 'reg__depth': 3, 'reg__learning_rate': 0.015218511742325067, 'reg__l2_leaf_reg': 34.39448411717762, 'reg__bagging_temperature': 3.6933466480486814, 'reg__random_strength': 4.930423446962909, 'reg__subsample': 0.7954214671691305, 'reg__min_data_in_leaf': 50, 'reg__leaf_estimation_iterations': 6, 'reg__rsm': 0.6454351127324762, 'reg__border_count': 83, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 5672.08682815434}

Rank 9: {'reg__iterations': 1000, 'reg__depth': 3, 'reg__learning_rate': 0.01491063307459948, 'reg__l2_leaf_reg': 30.016958519515164, 'reg__bagging_temperature': 3.618331973529094, 'reg__random_strength': 4.953487746086349, 'reg__subsample': 0.7877630961361051, 'reg__min_data_in_leaf': 40, 'reg__leaf_estimation_iterations': 5, 'reg__rsm': 0.6457521886625761, 'reg__border_count': 87, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 7865.512565455419}

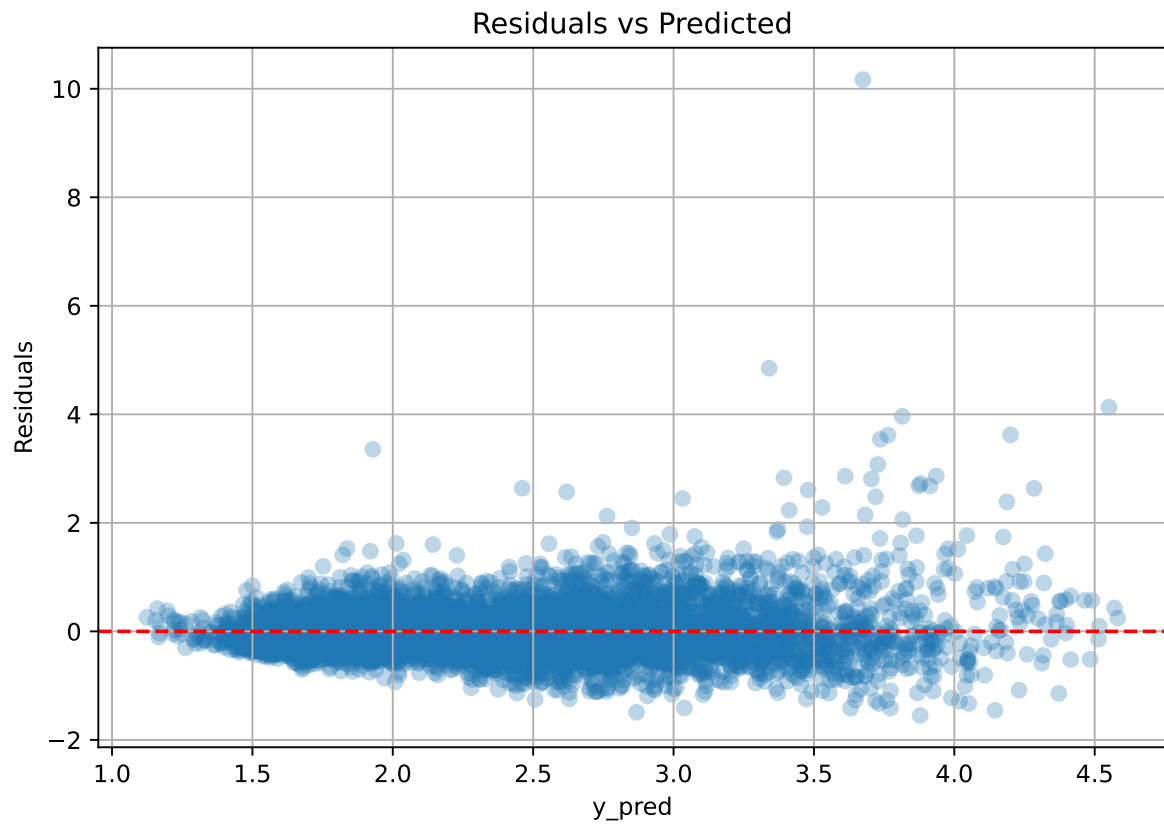
Rank 10: {'reg__iterations': 800, 'reg__depth': 3, 'reg__learning_rate': 0.018068985853702778, 'reg__l2_leaf_reg': 19.401459017460358, 'reg__bagging_temperature': 3.942018859630849,

```
'reg__random_strength': 4.531052499947106, 'reg__subsample': 0.6239368911224855,  
'reg__min_data_in_leaf': 41, 'reg__leaf_estimation_iterations': 6, 'reg__rsm':  
0.6586489766247147, 'reg__border_count': 73, 'reg__leaf_estimation_method': 'Gradient',  
'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_tempera-  
ture': 9667.9927670665}
```

3.6.2 Scatter



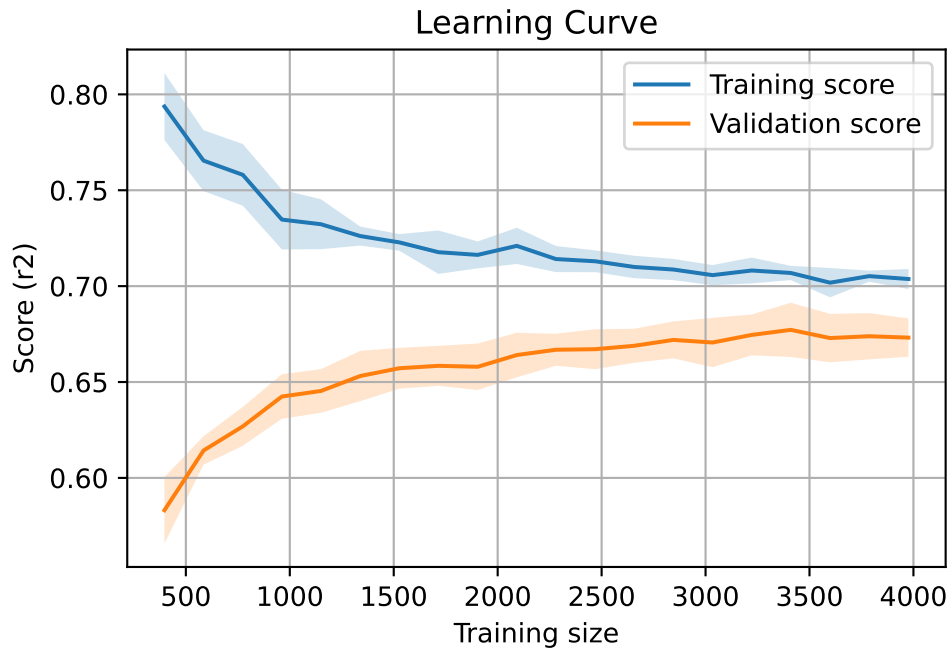
3.6.3 Scatter residuals



3.6.4 Learning curve

`Len(group)` : 8116, `Len(training)` : 5682,

`Len(training_real)` : 3978, `Len(test_of_training)` : 1704, `Len(test)` : 2434



3.7 Sumup-table

model	mean_train_cv	mean_test_cv	std_test_cv	test_cv	test	diff	r_test	r_train	pen_score
Linear	0.534	0.529	0.013	0.503	0.504	-0.025	0.953	1.01	nan
LGBM	0.708	0.675	0.012	0.631	0.646	-0.029	0.957	1.049	0.6752
CatBoost	0.704	0.672	0.011	0.631	0.641	-0.031	0.954	1.046	0.6723

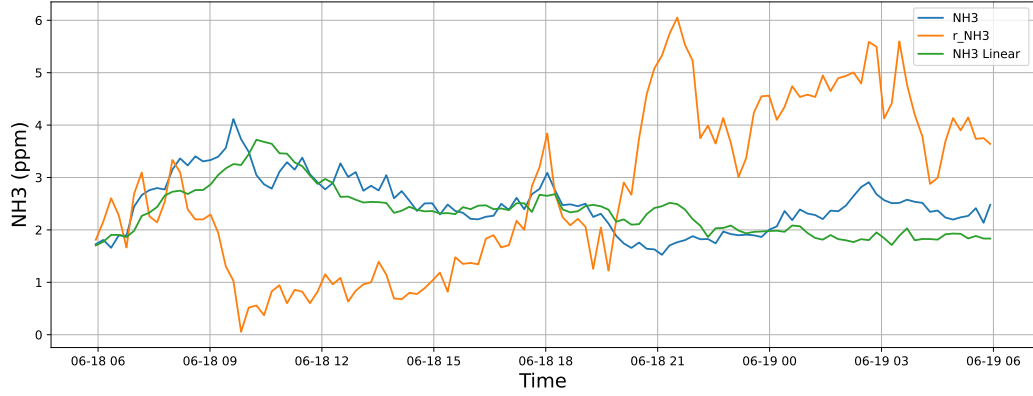
4 Plot

4.1 Standalone plots

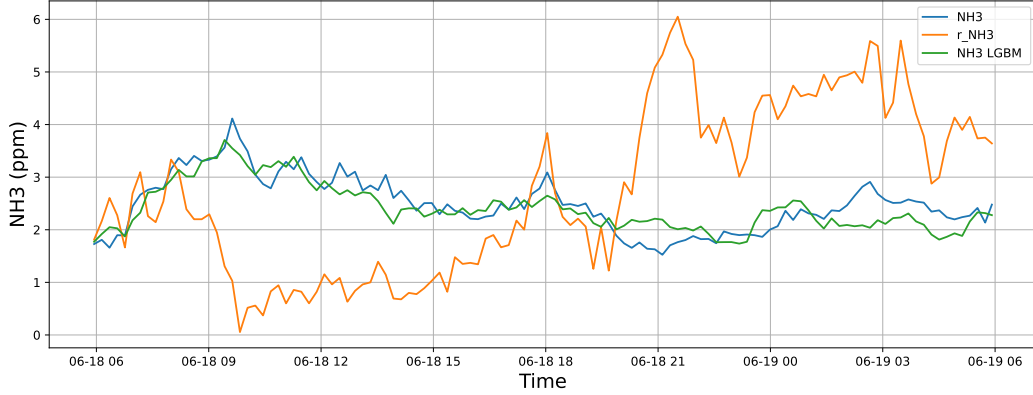
4.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

4.1.1.1 Time window : 24

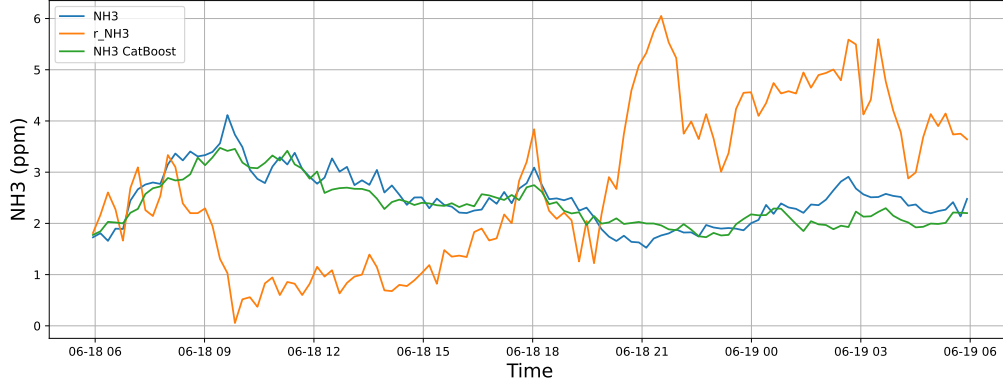
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

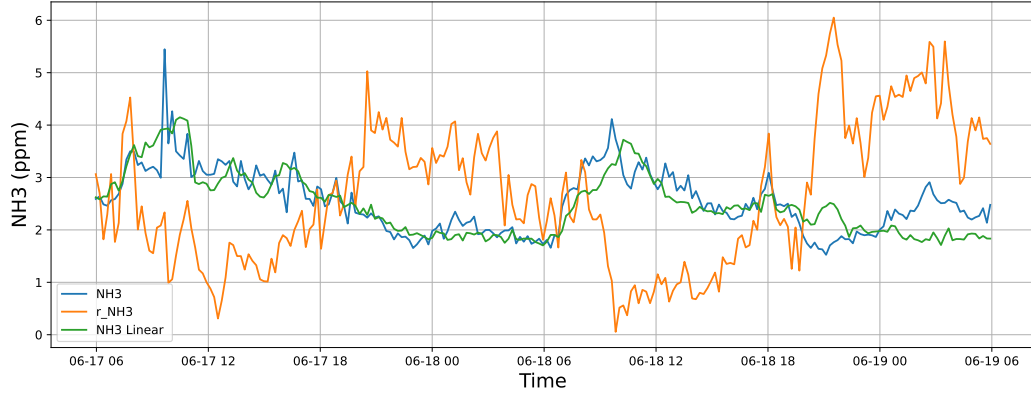


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

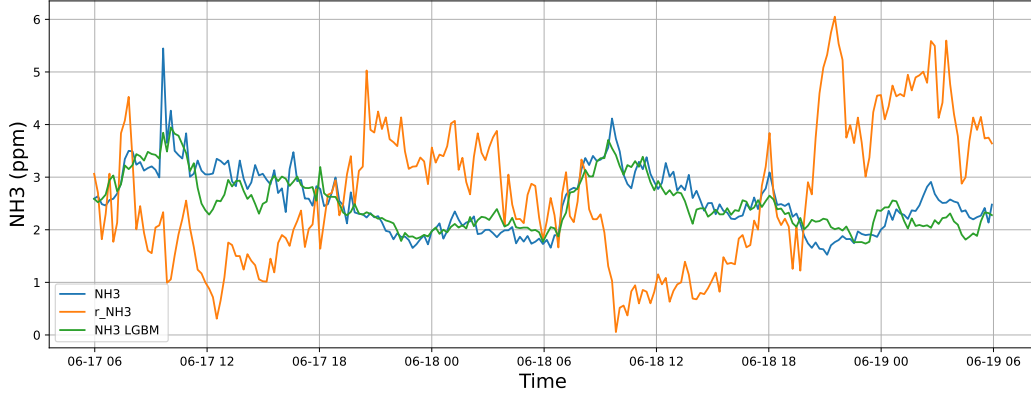


4.1.1.2 Time window : 48

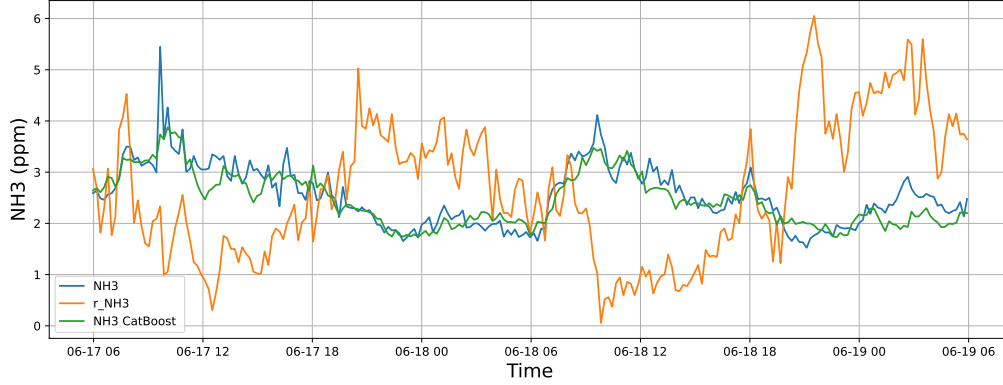
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

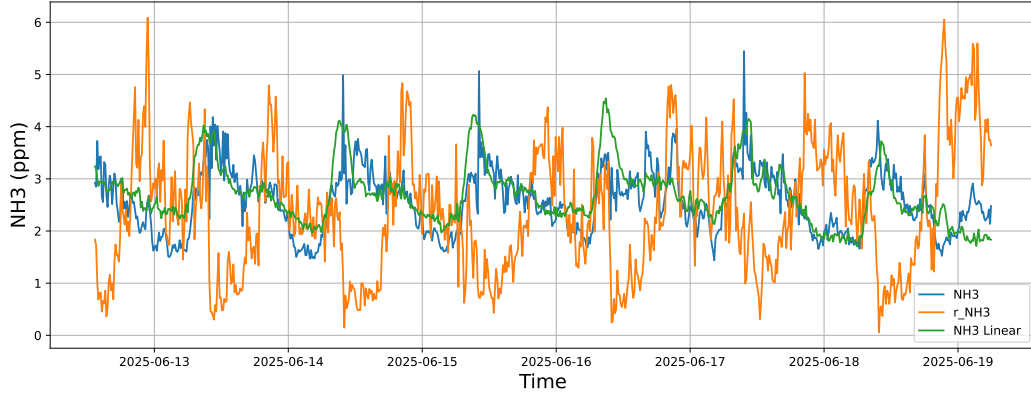


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

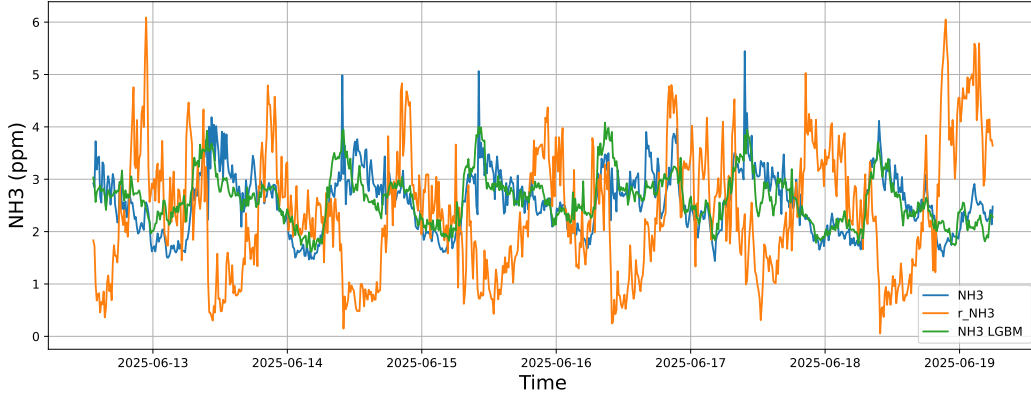


4.1.1.3 Time window : ALL

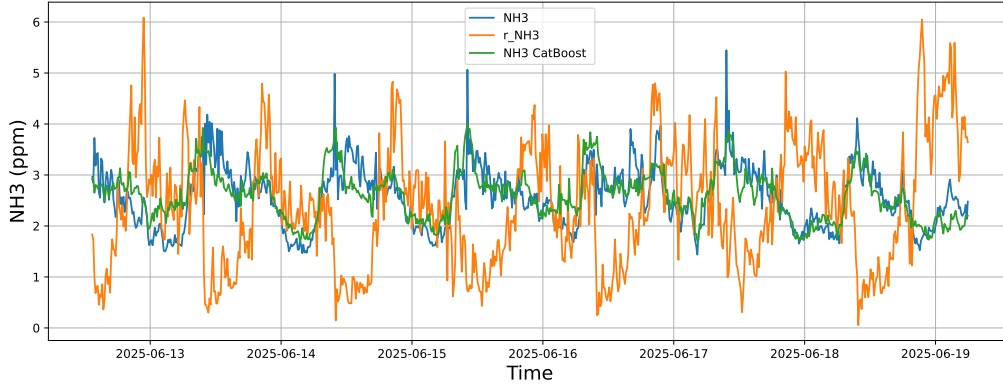
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



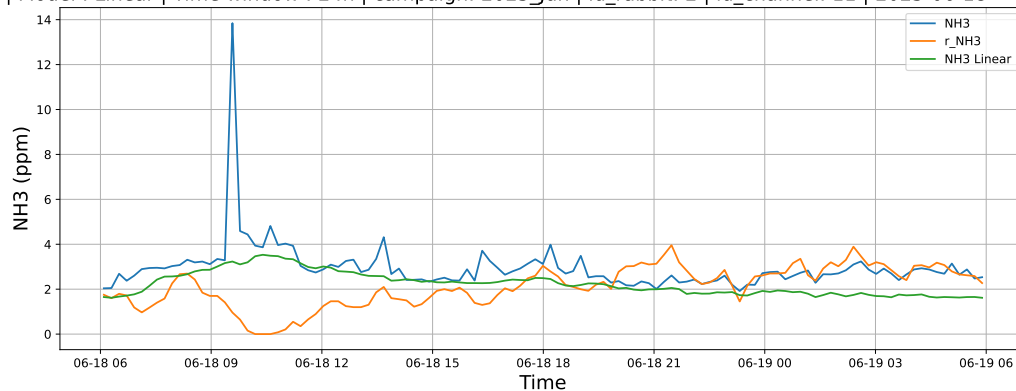
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



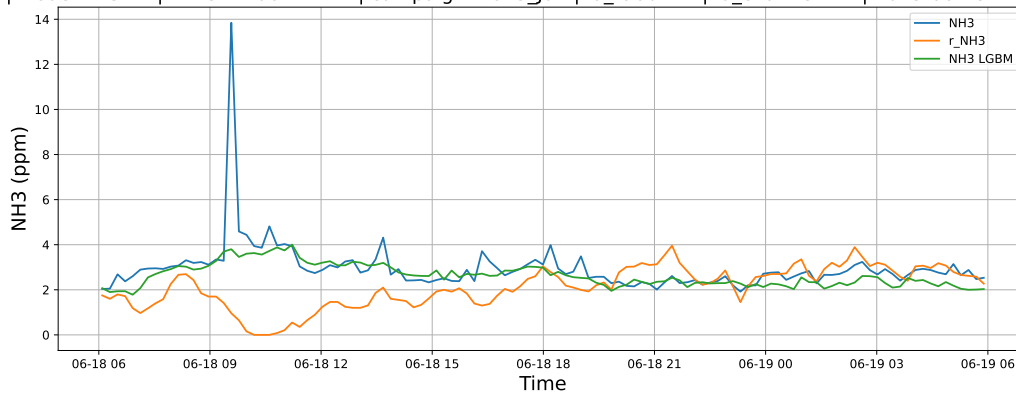
4.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

4.1.2.1 Time window : 24

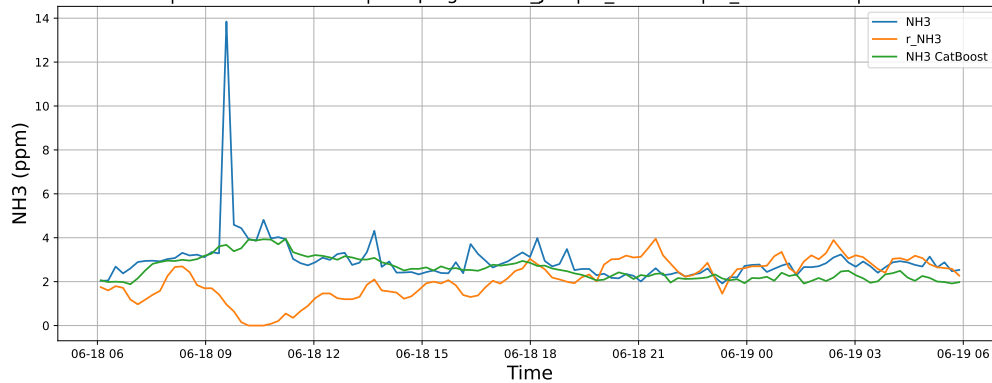
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

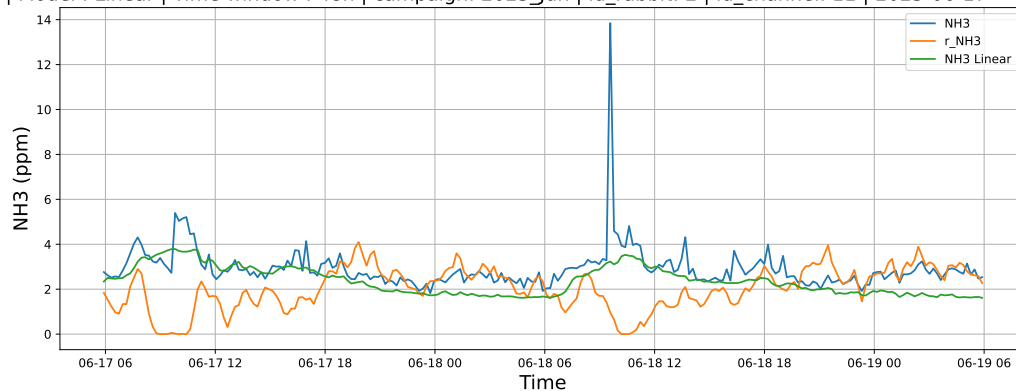


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

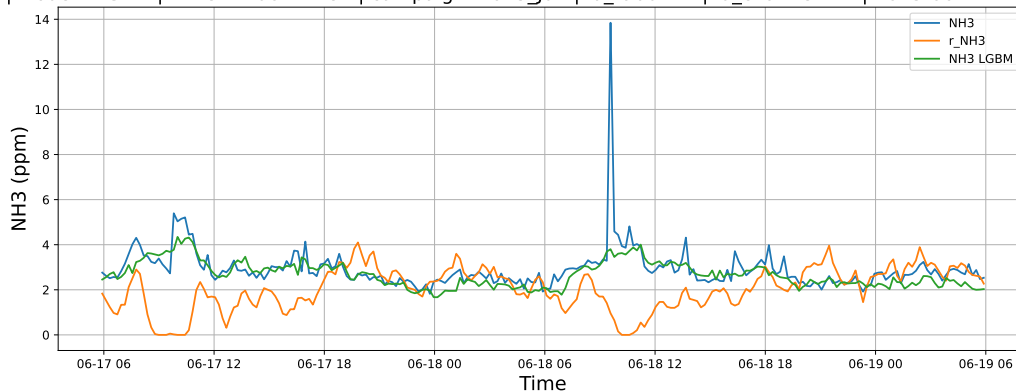


4.1.2.2 Time window : 48

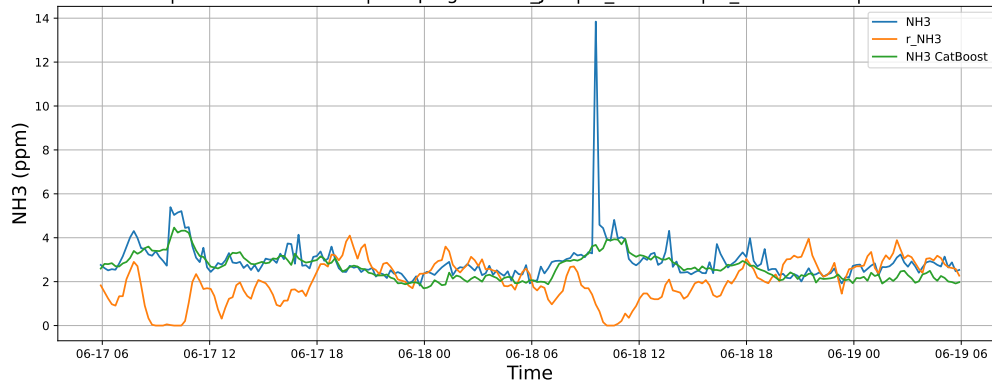
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

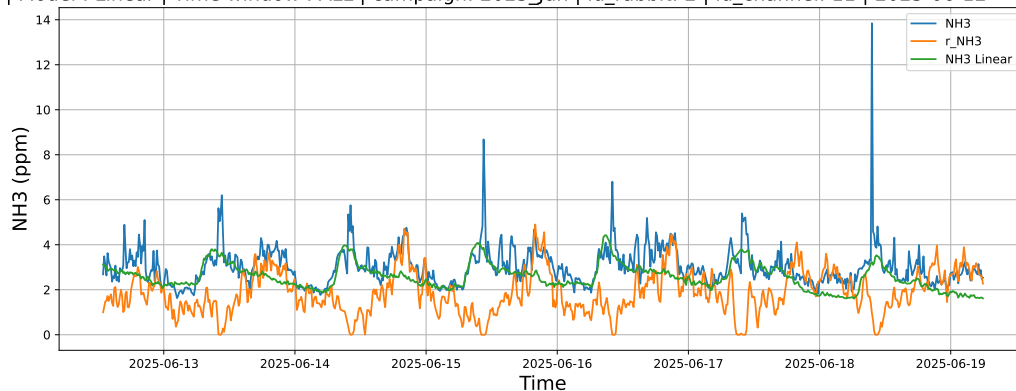


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

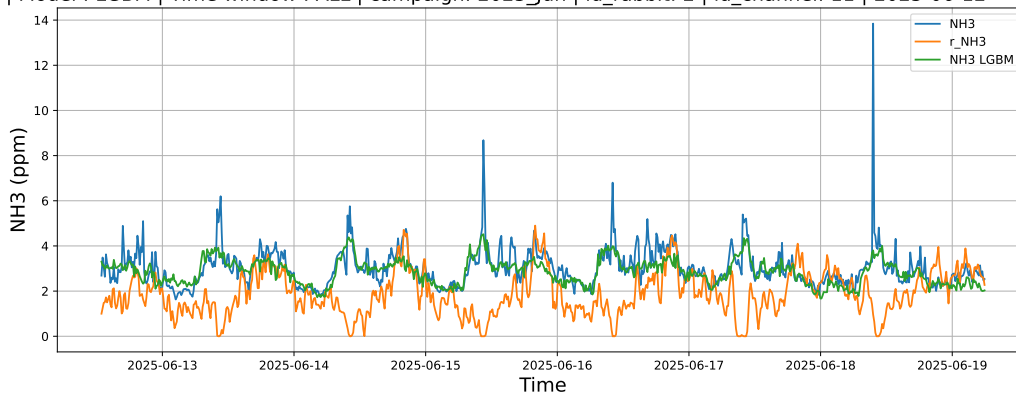


4.1.2.3 Time window : ALL

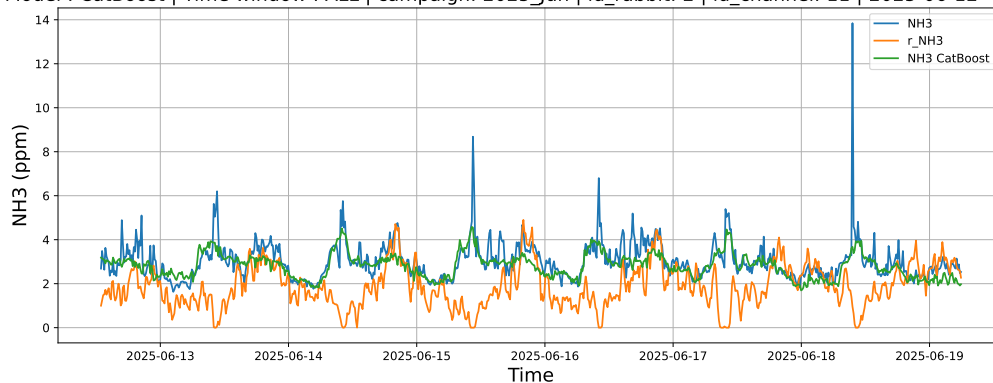
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



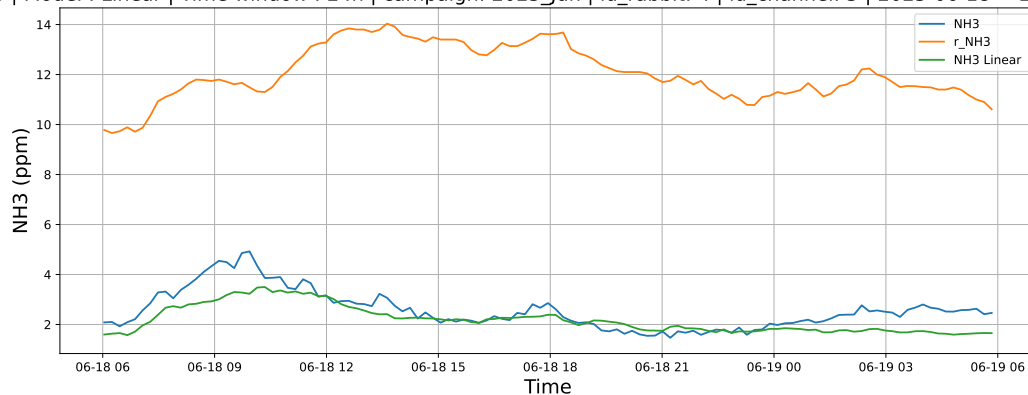
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



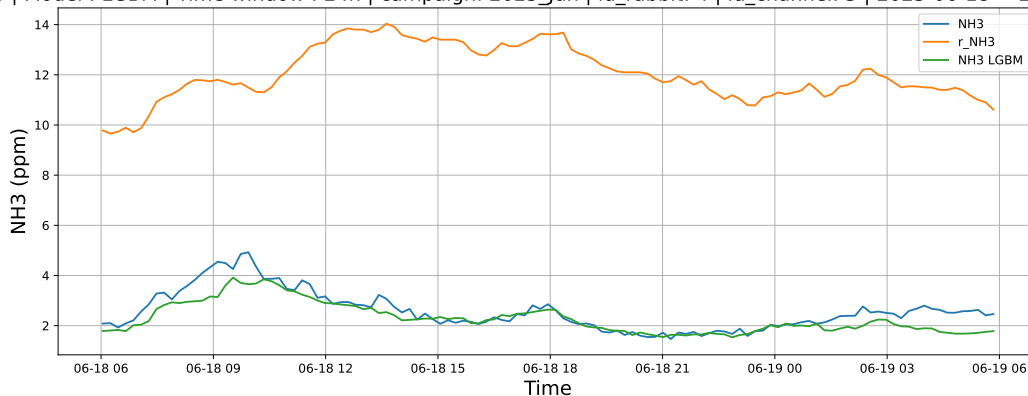
4.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

4.1.3.1 Time window : 24

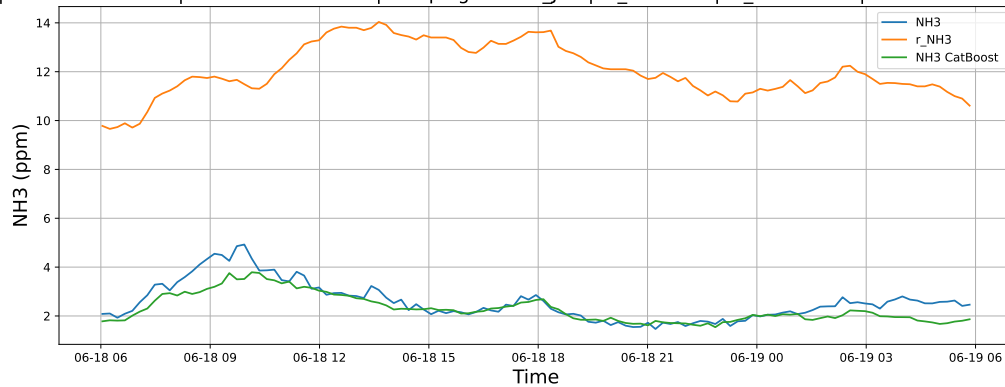
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

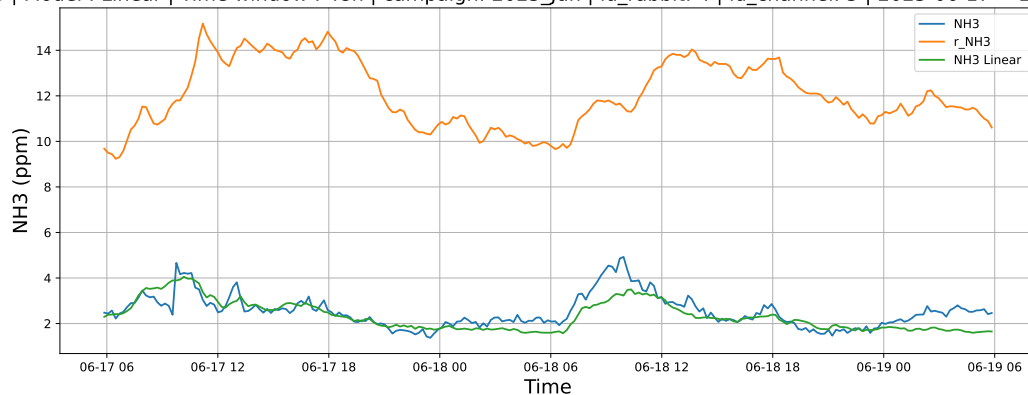


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

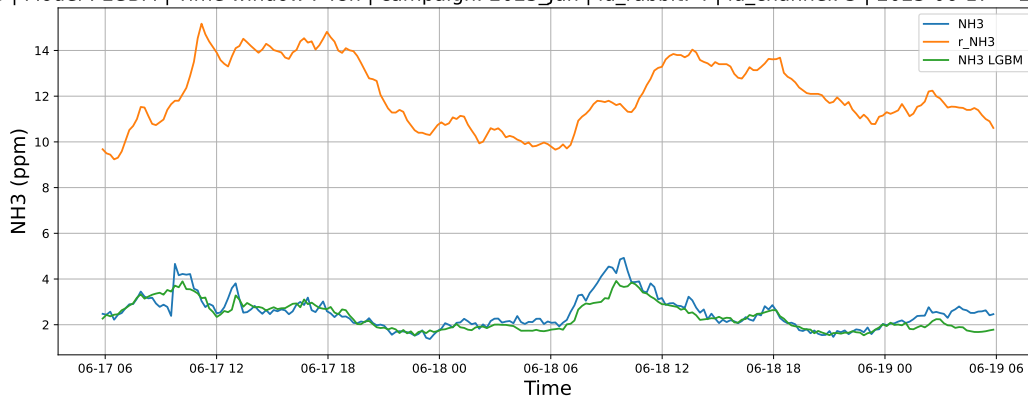


4.1.3.2 Time window : 48

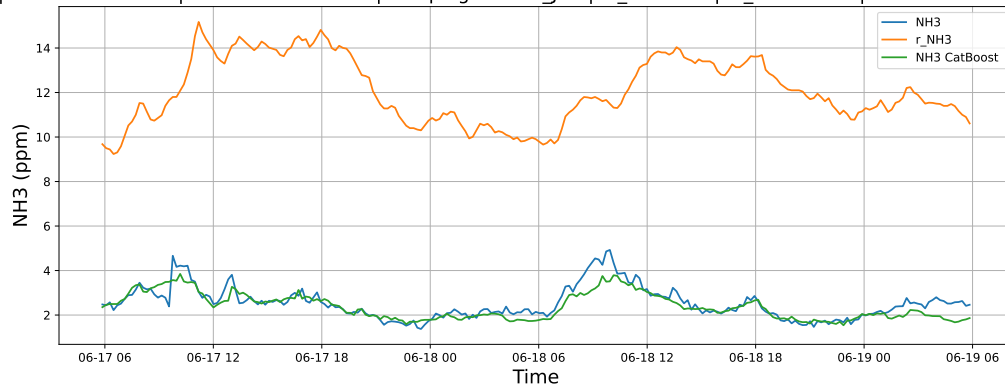
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

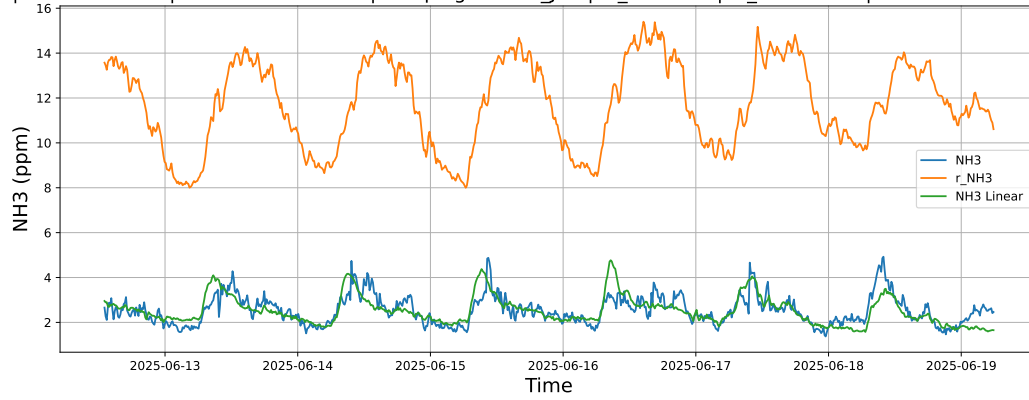


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

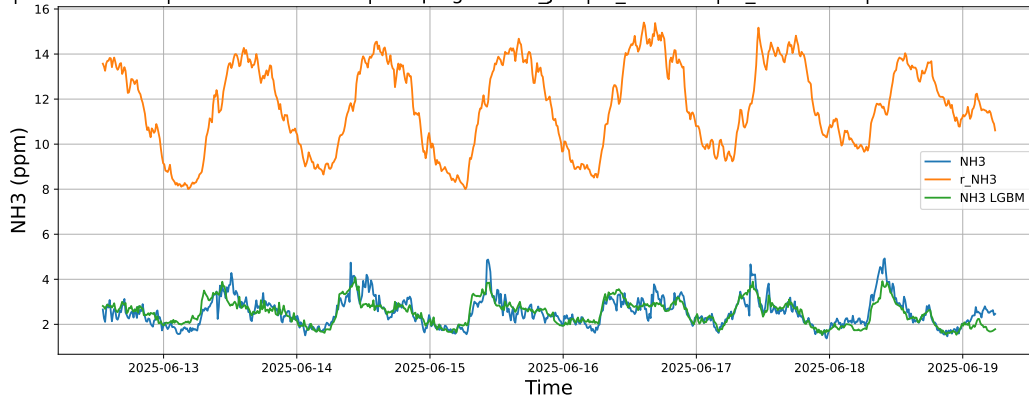


4.1.3.3 Time window : ALL

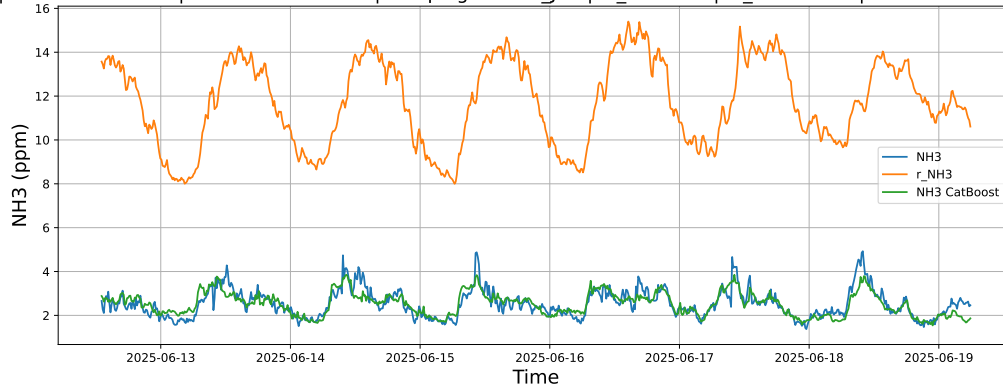
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



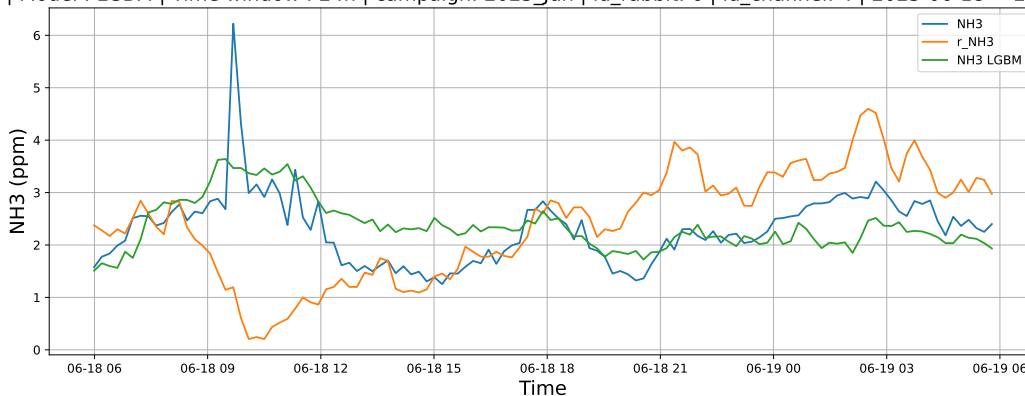
4.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

4.1.4.1 Time window : 24

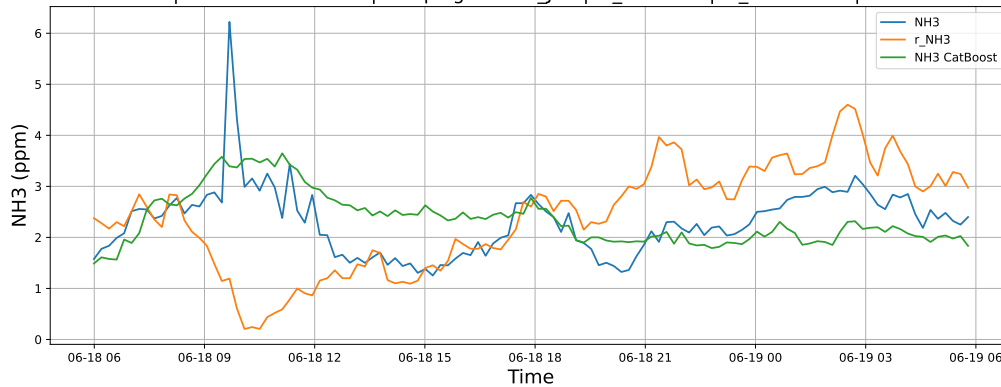
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

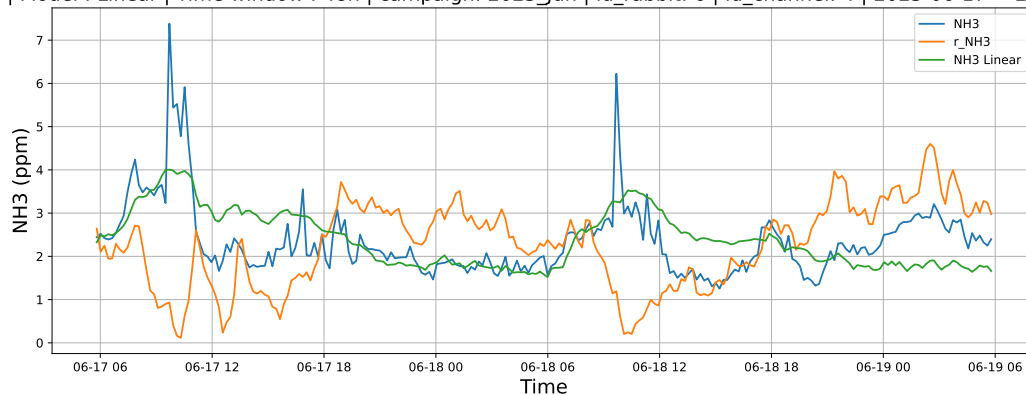


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

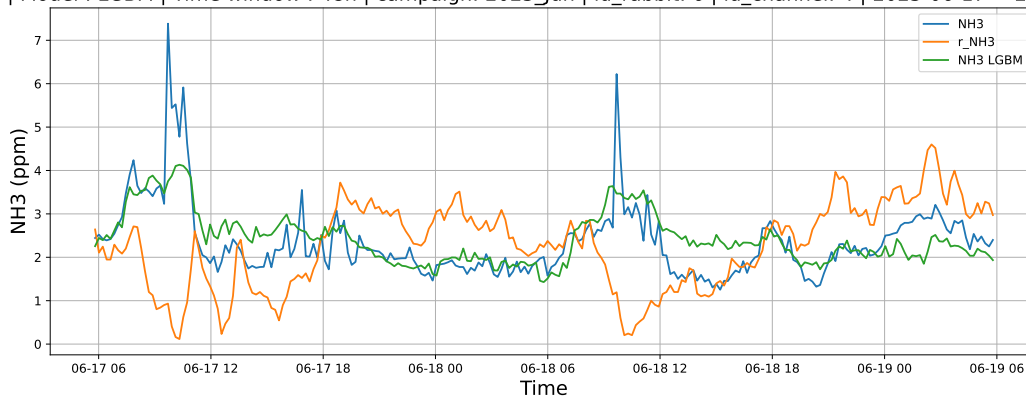


4.1.4.2 Time window : 48

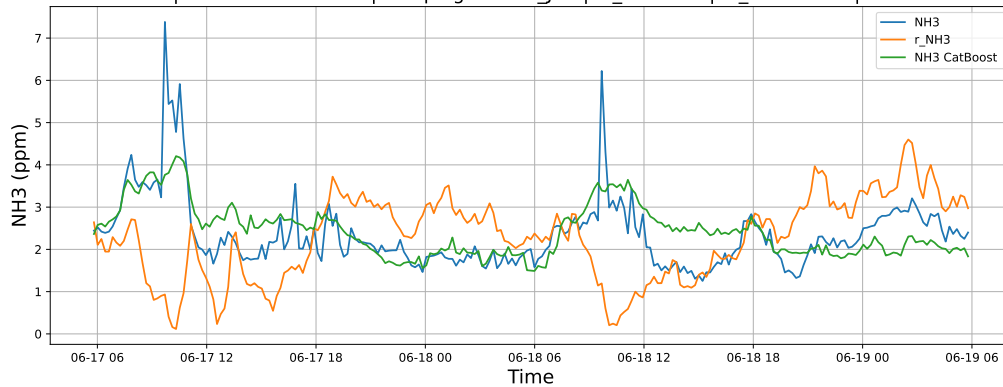
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

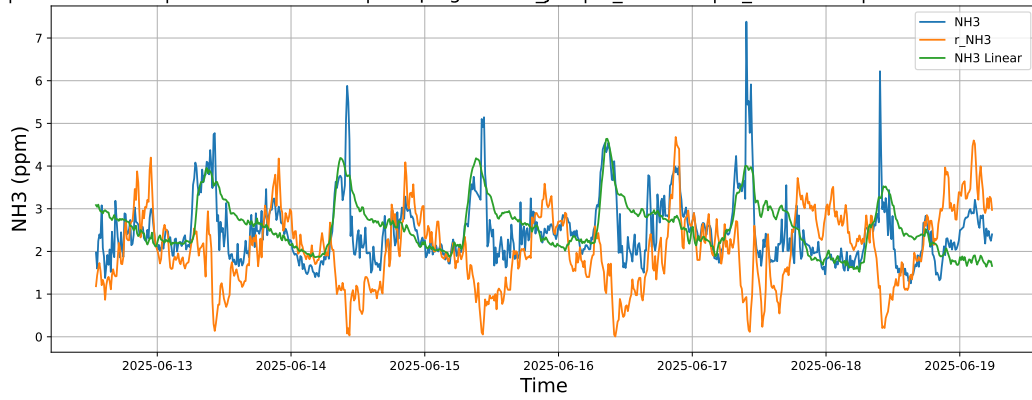


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

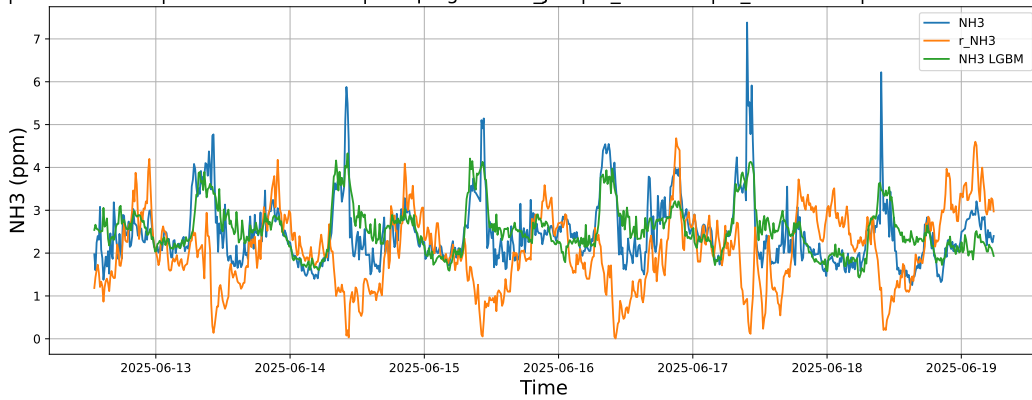


4.1.4.3 Time window : ALL

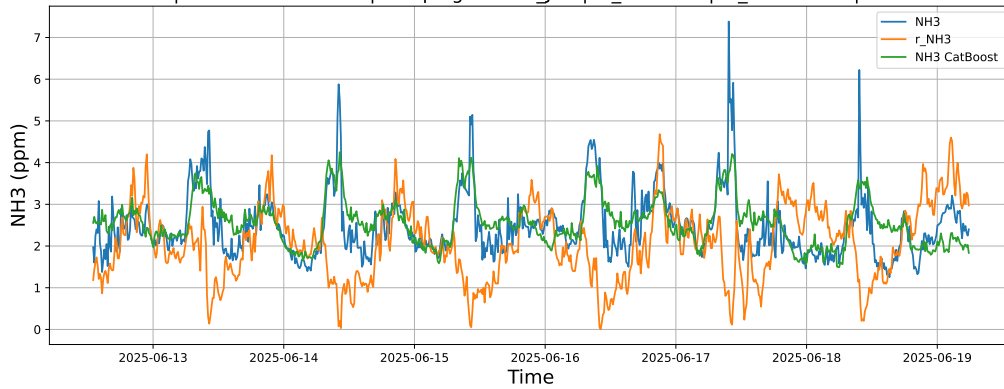
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



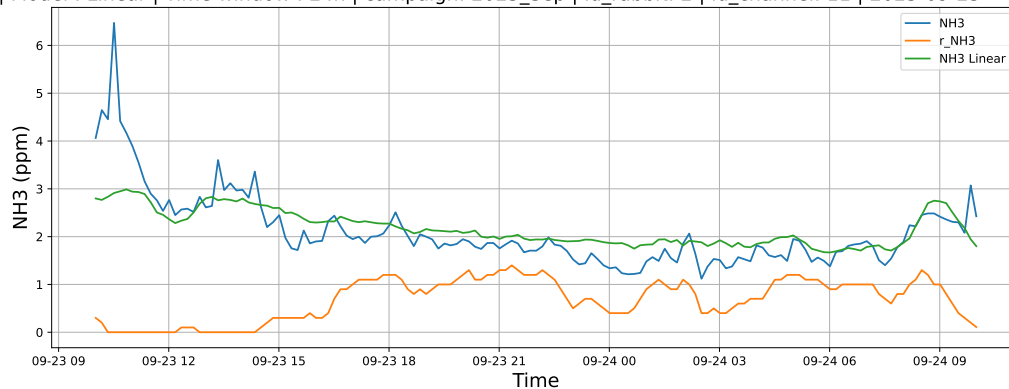
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



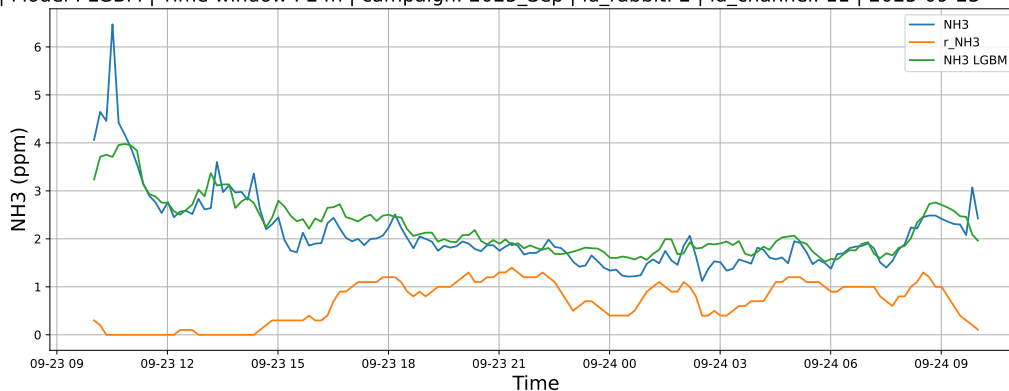
4.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

4.1.5.1 Time window : 24

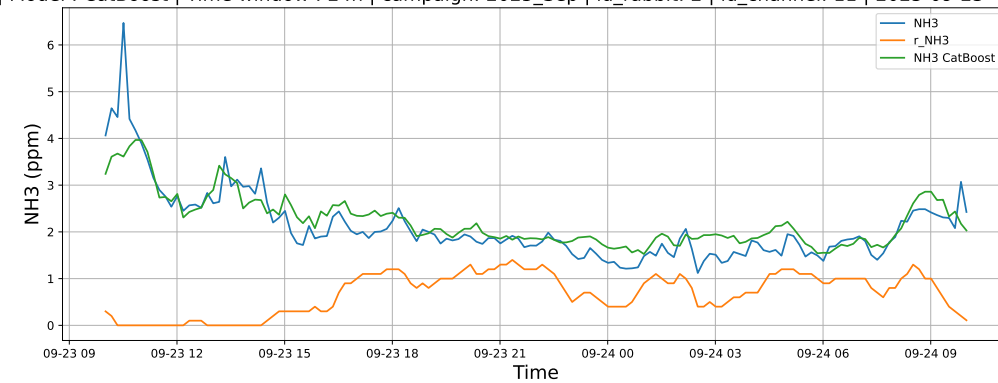
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

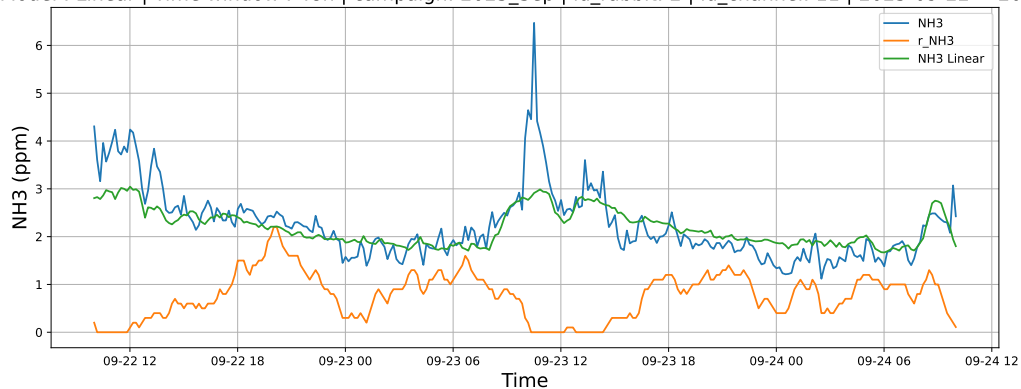


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

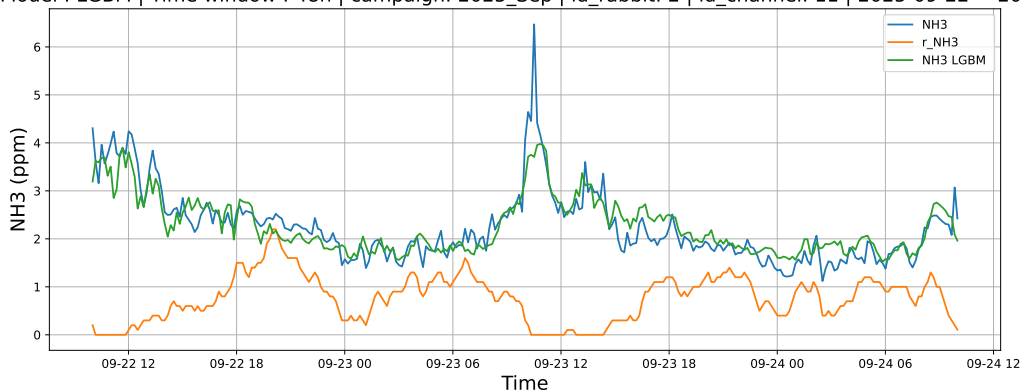


4.1.5.2 Time window : 48

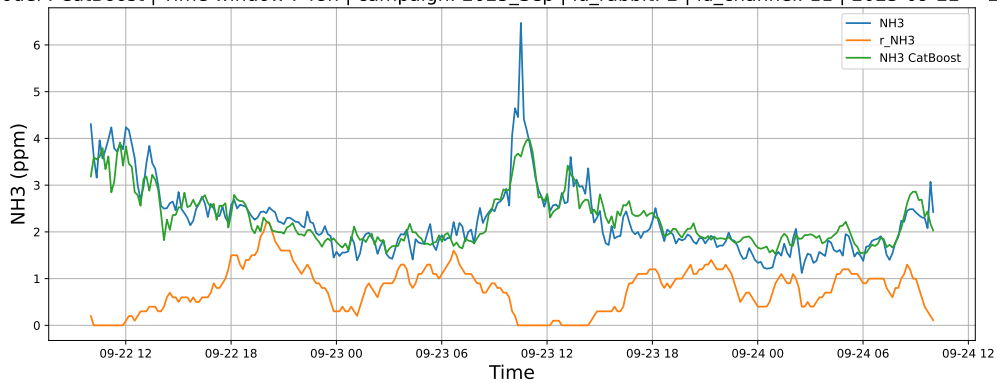
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

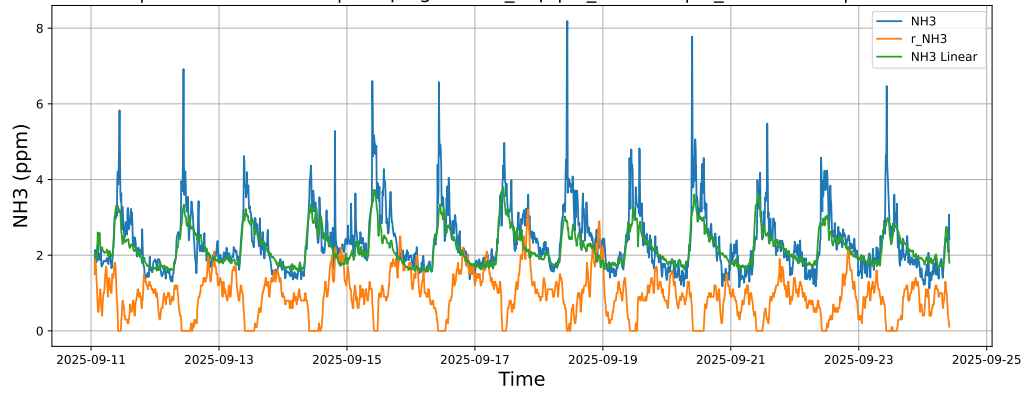


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

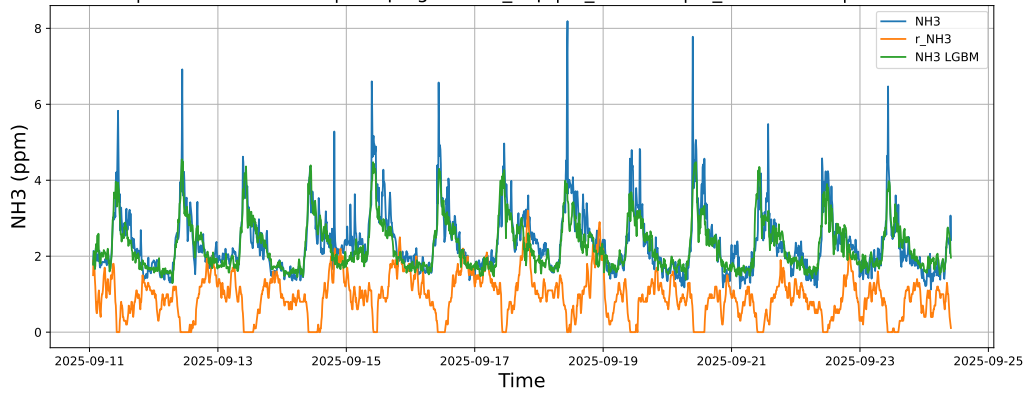


4.1.5.3 Time window : ALL

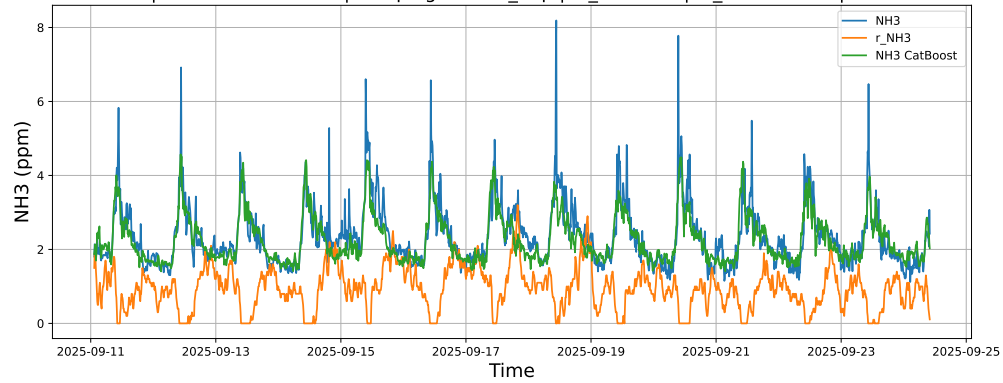
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



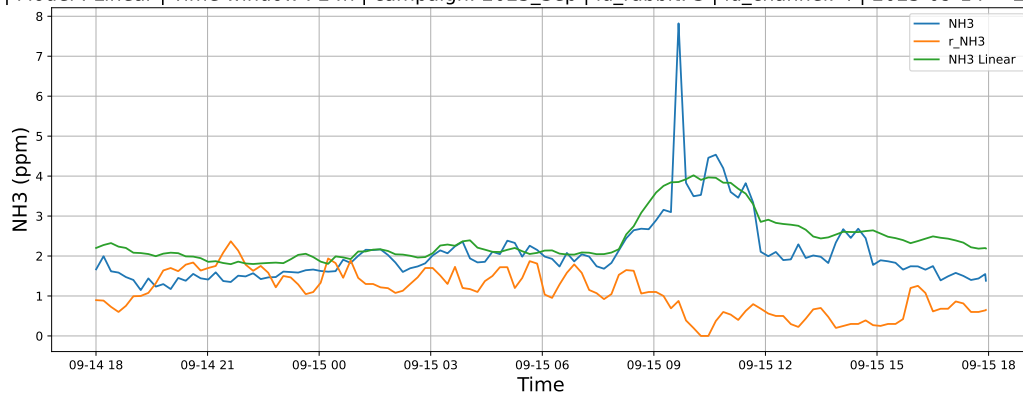
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



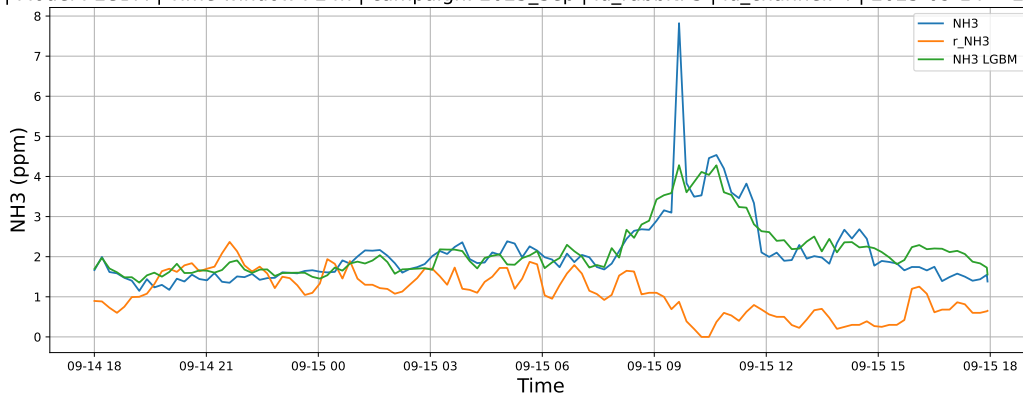
4.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

4.1.6.1 Time window : 24

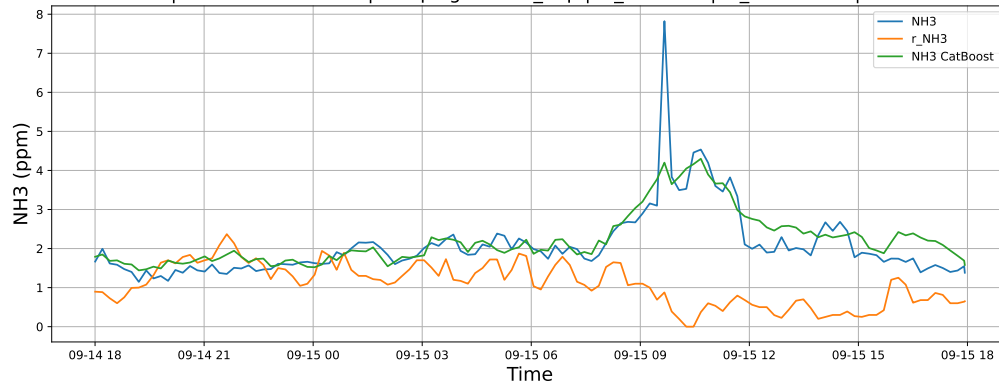
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

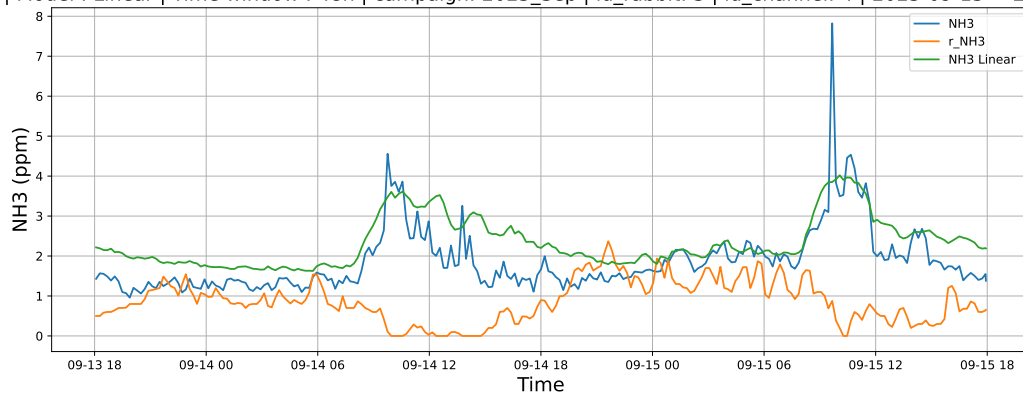


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

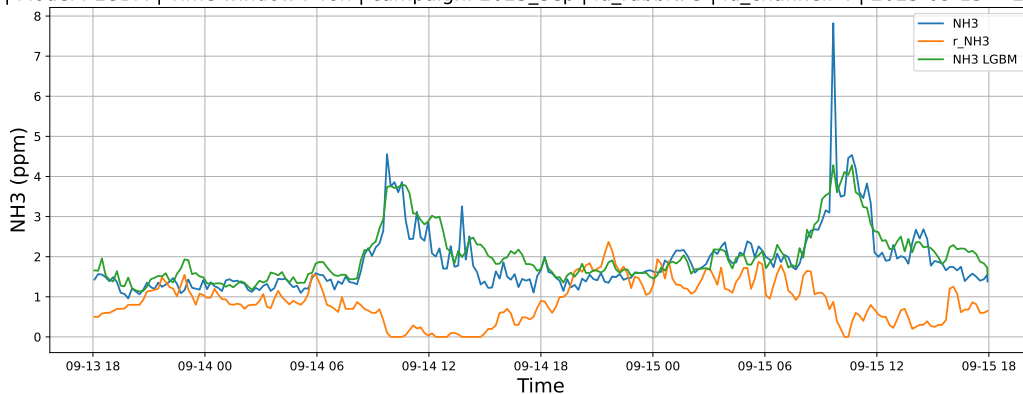


4.1.6.2 Time window : 48

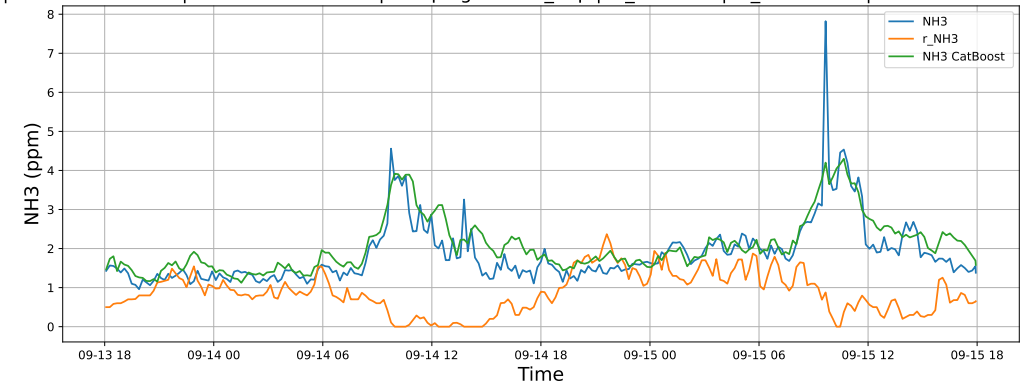
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

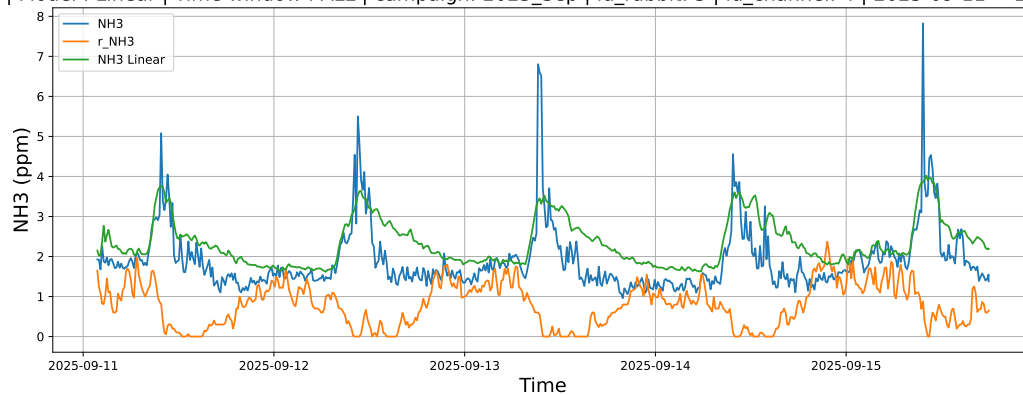


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

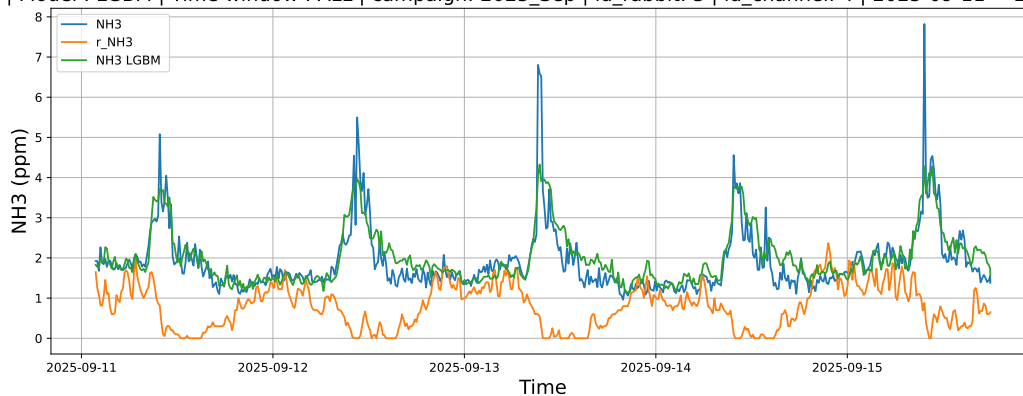


4.1.6.3 Time window : ALL

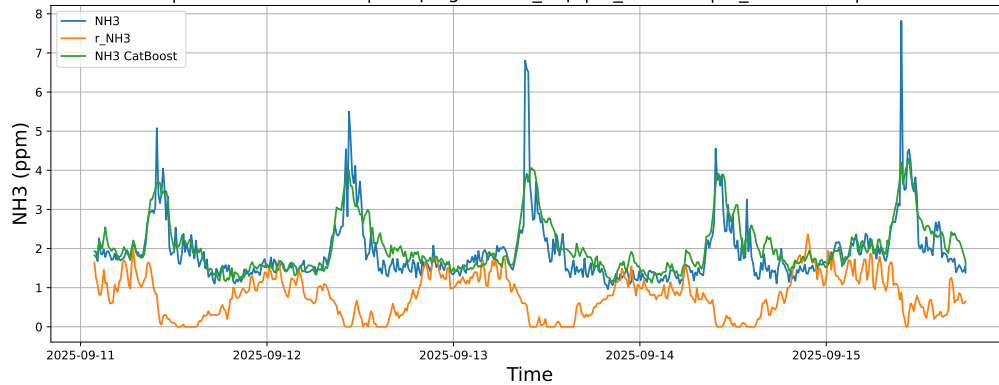
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



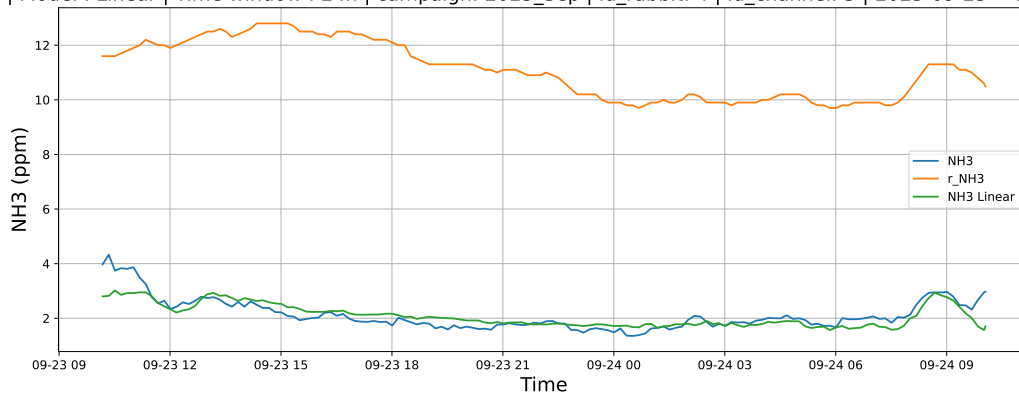
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



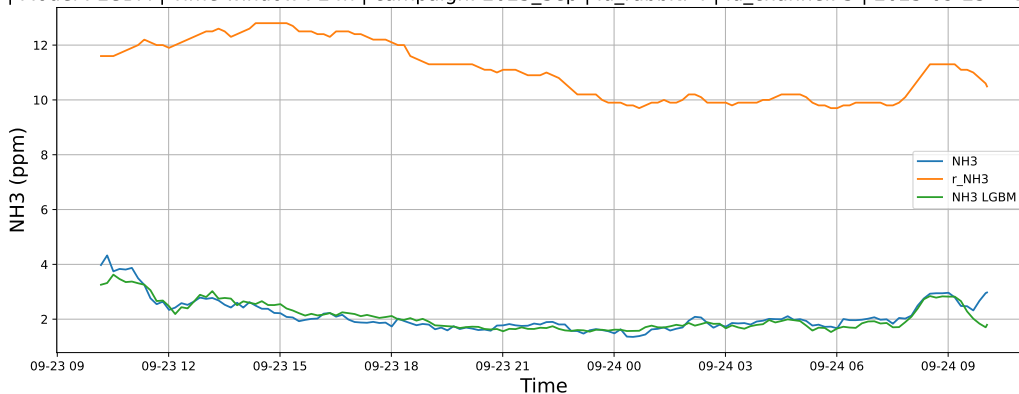
4.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

4.1.7.1 Time window : 24

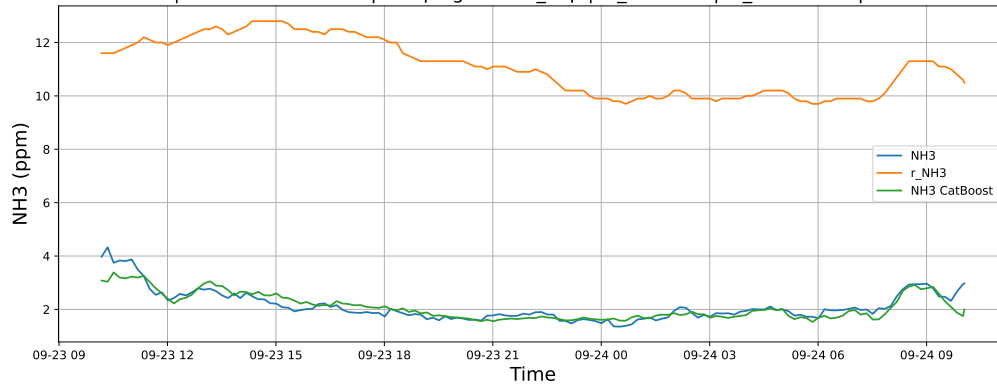
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

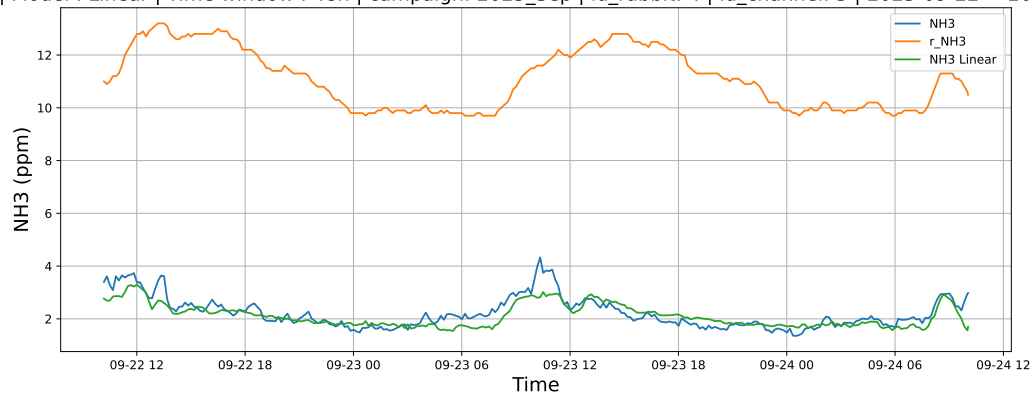


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

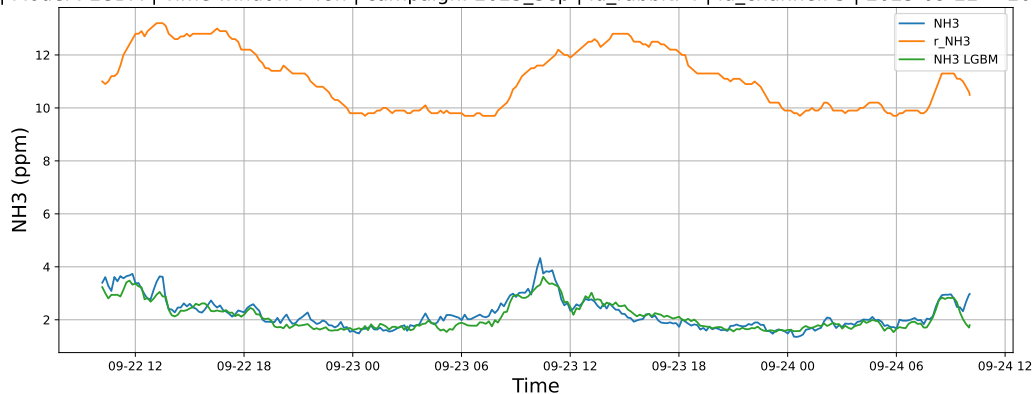


4.1.7.2 Time window : 48

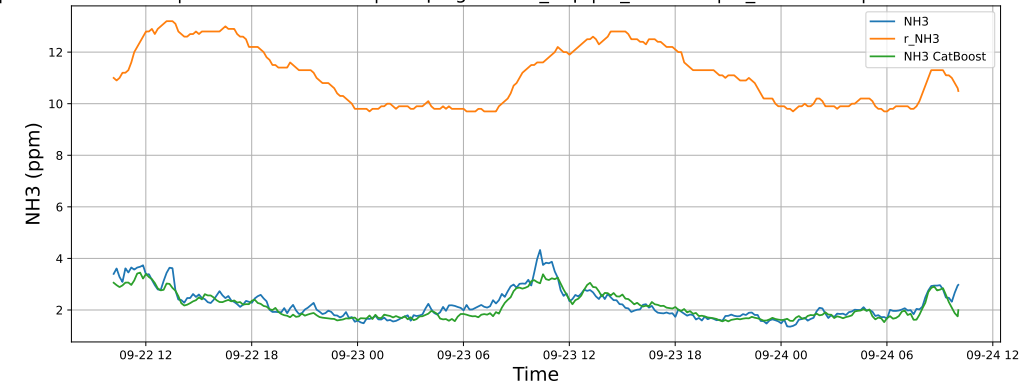
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

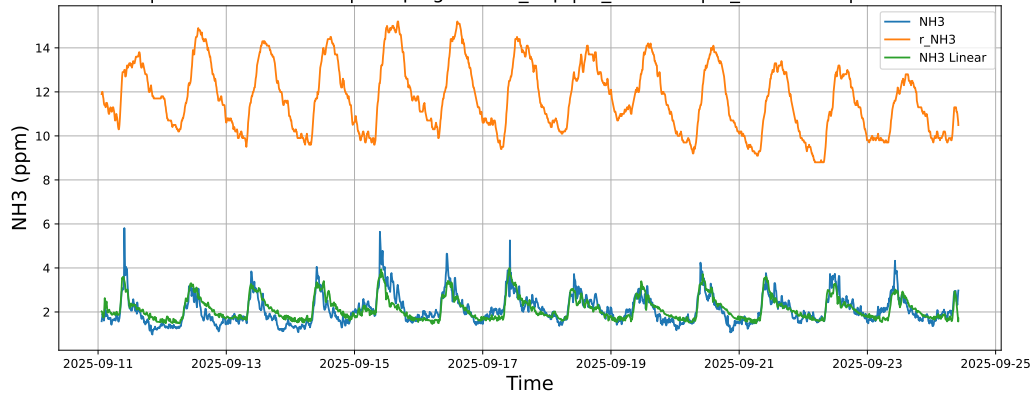


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

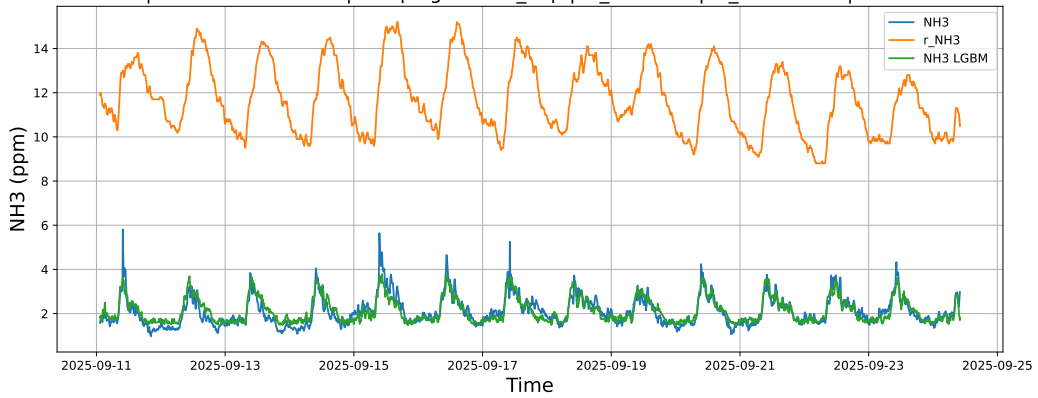


4.1.7.3 Time window : ALL

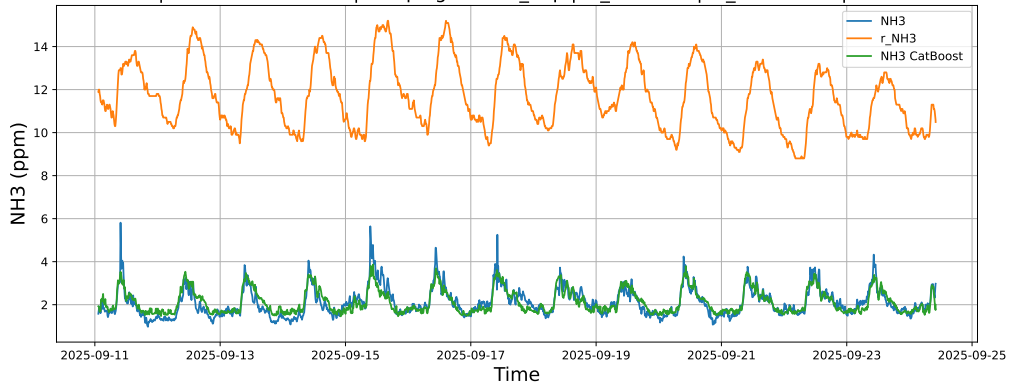
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



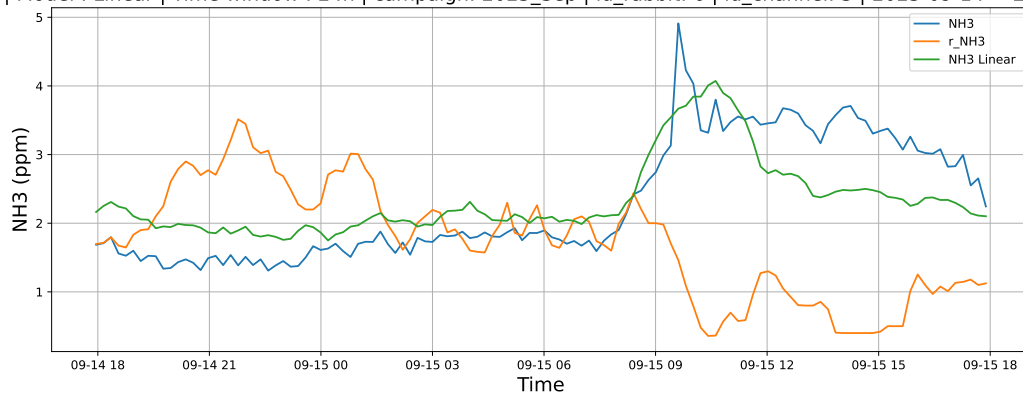
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



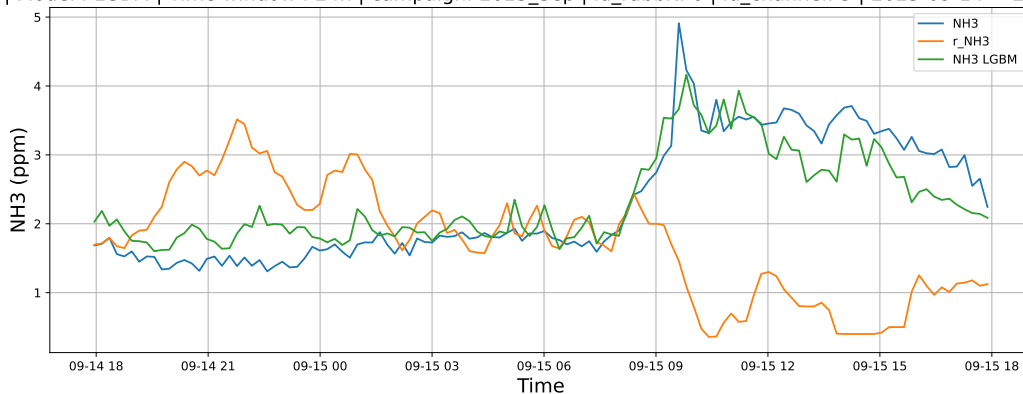
4.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

4.1.8.1 Time window : 24

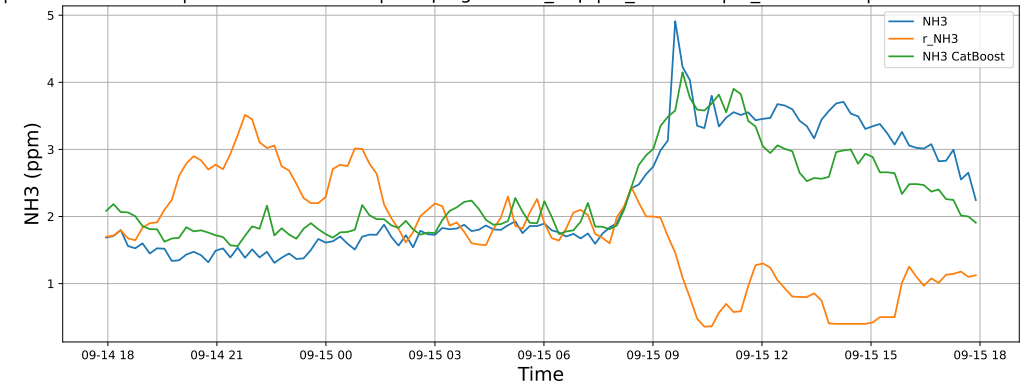
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

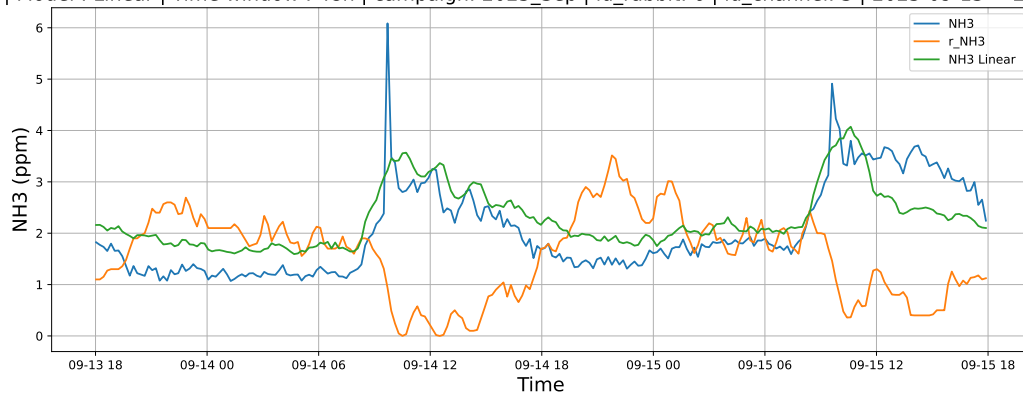


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

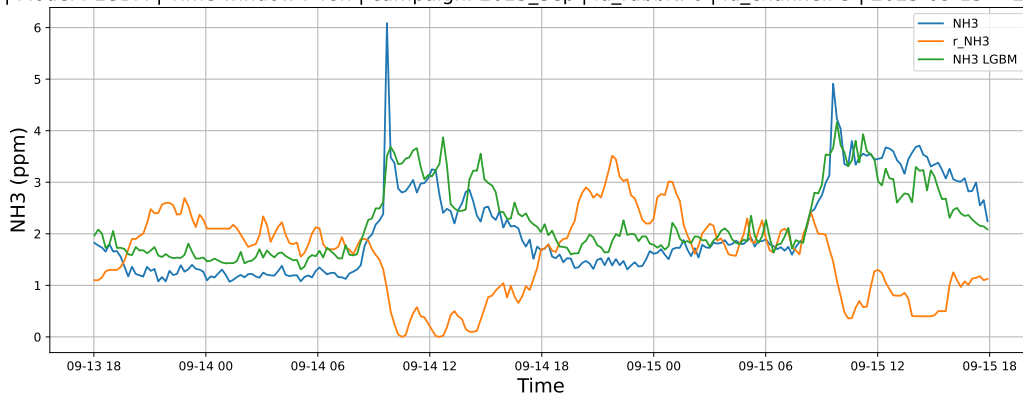


4.1.8.2 Time window : 48

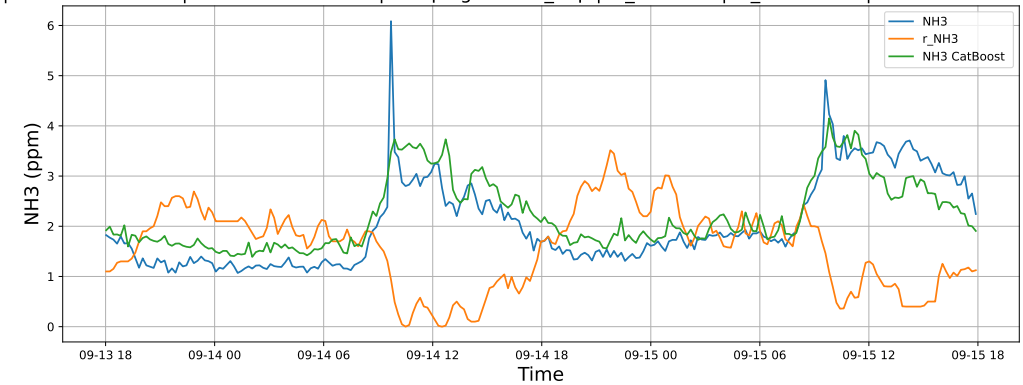
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

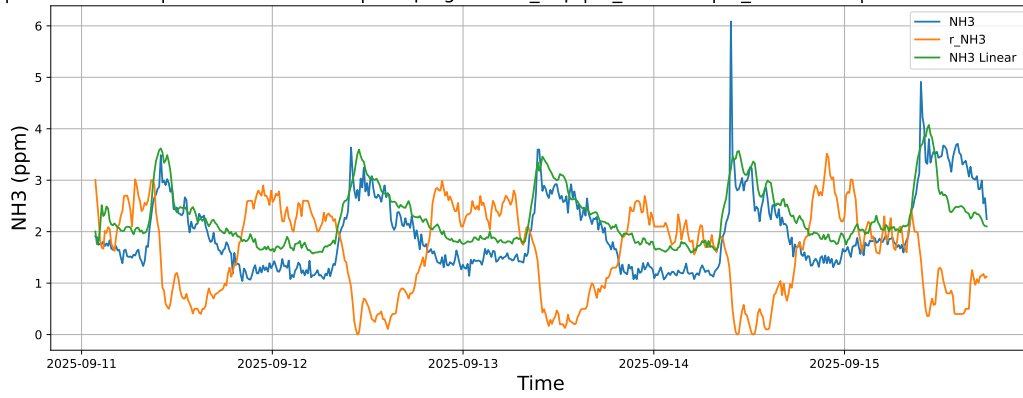


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

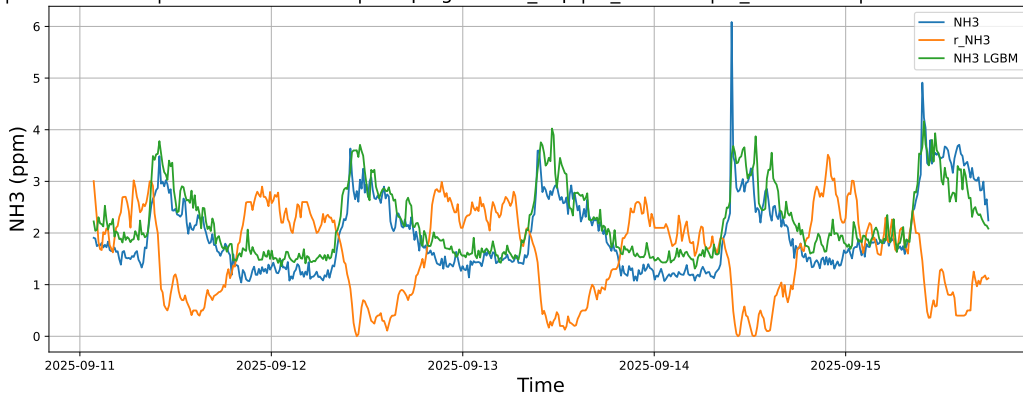


4.1.8.3 Time window : ALL

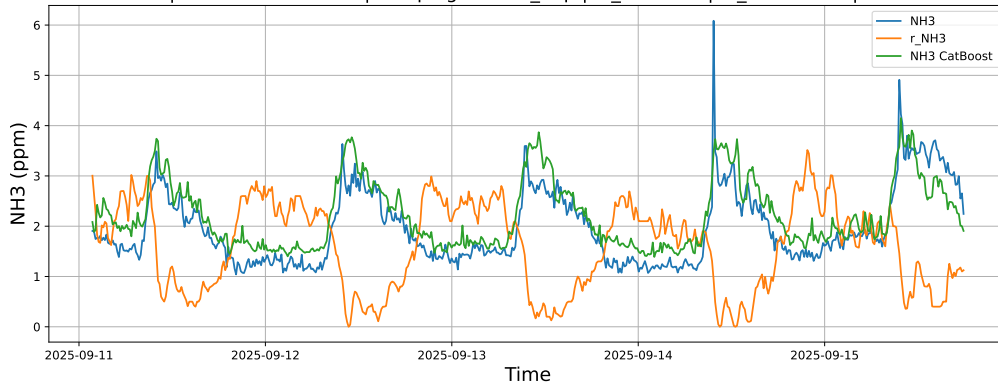
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

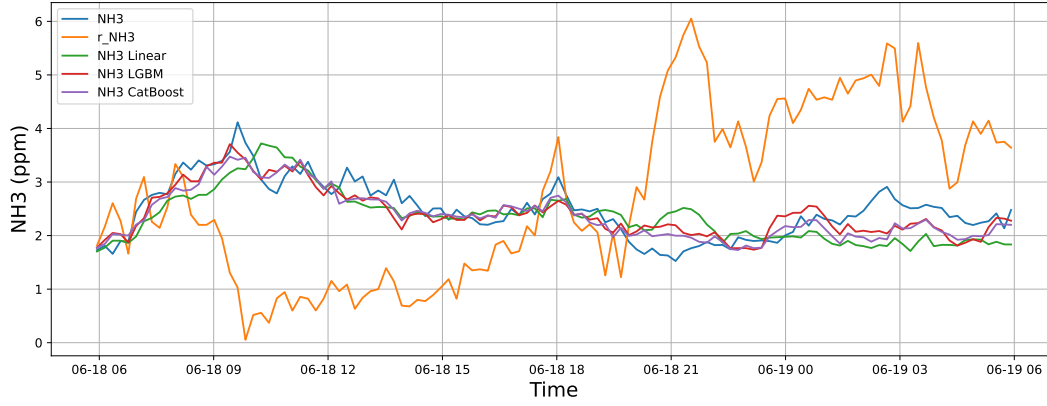


4.2 Compare plots

4.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

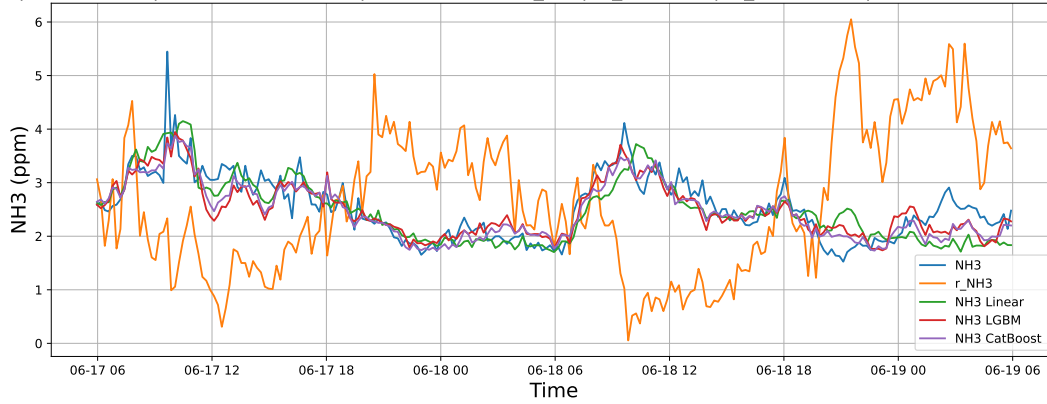
4.2.1.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



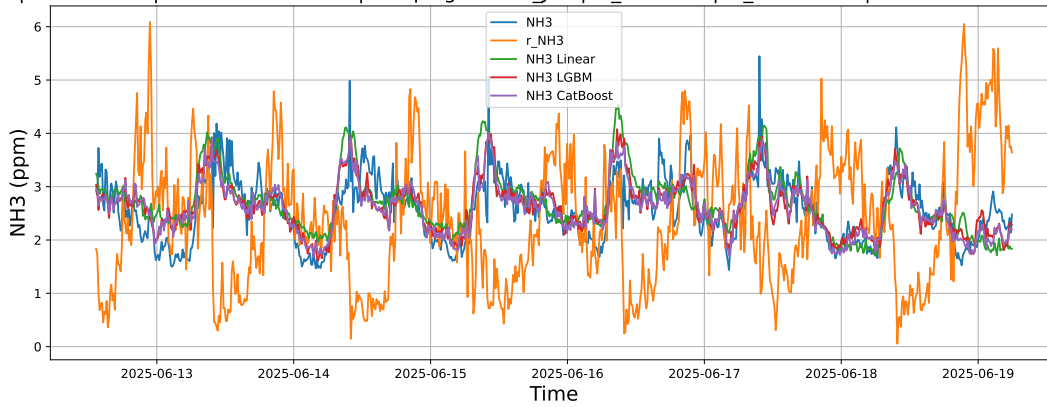
4.2.1.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



4.2.1.3 Time window : ALL

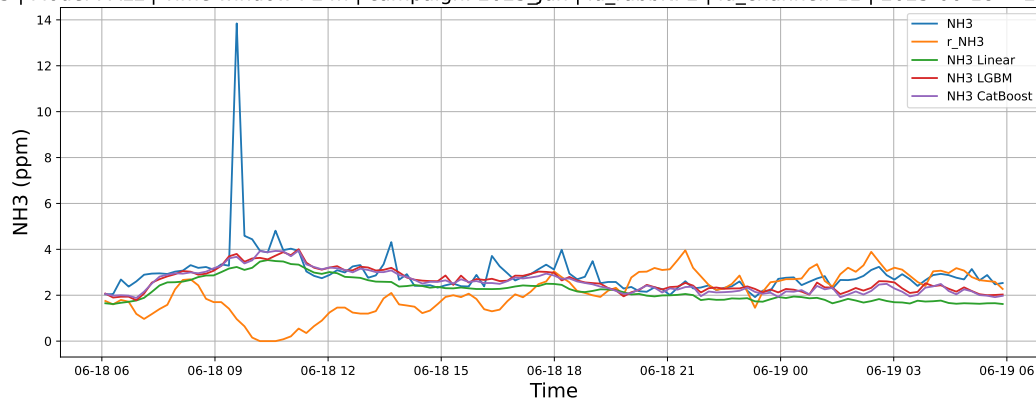
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



4.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

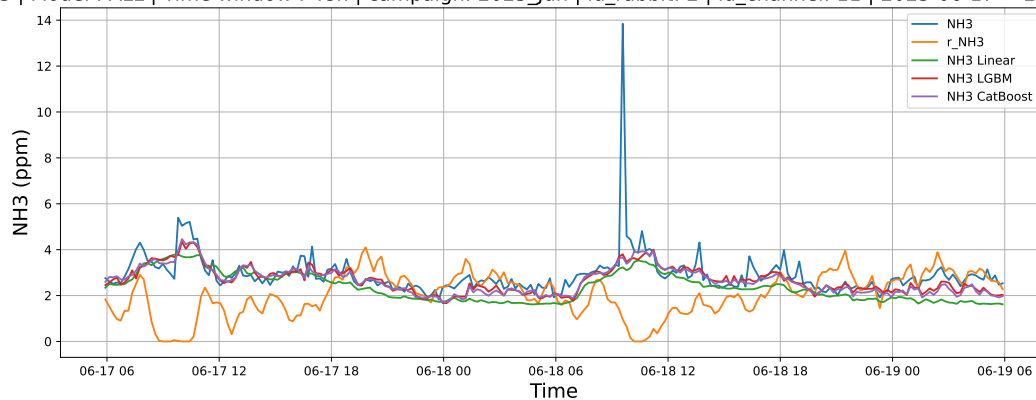
4.2.2.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



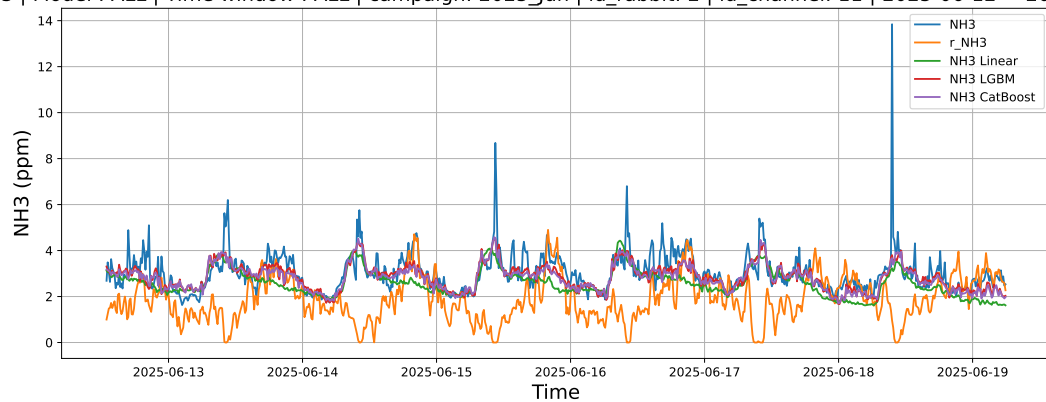
4.2.2.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



4.2.2.3 Time window : ALL

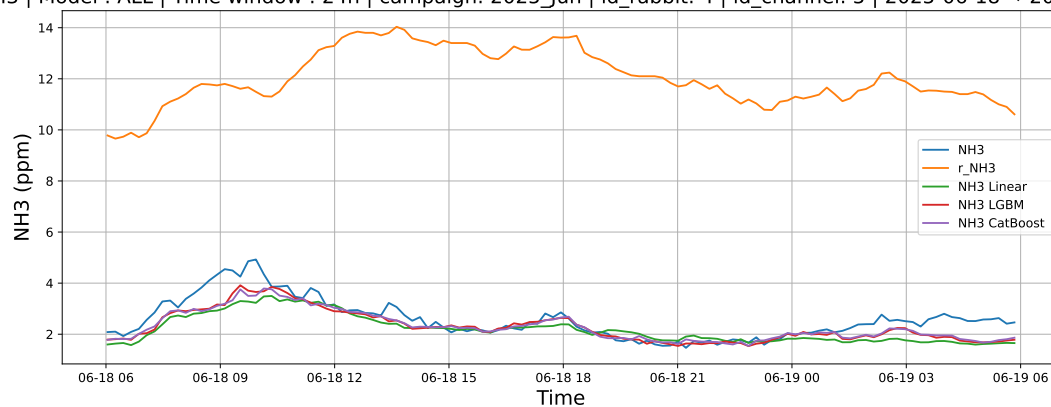
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



4.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

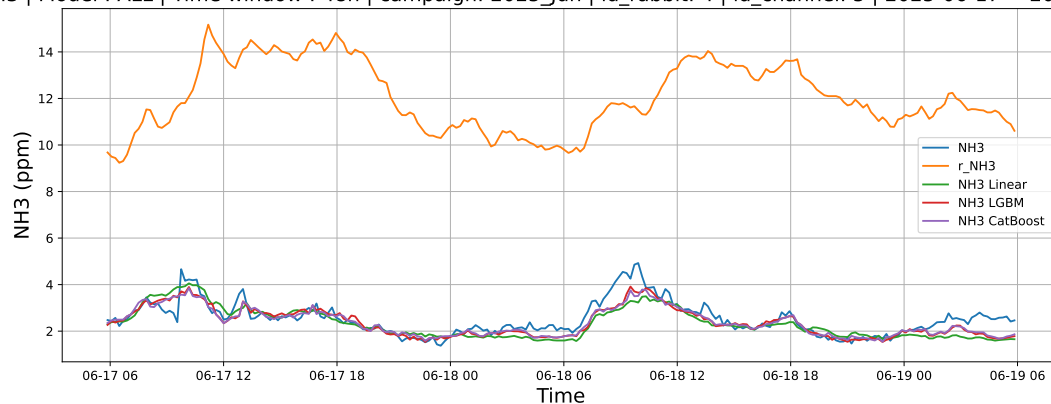
4.2.3.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



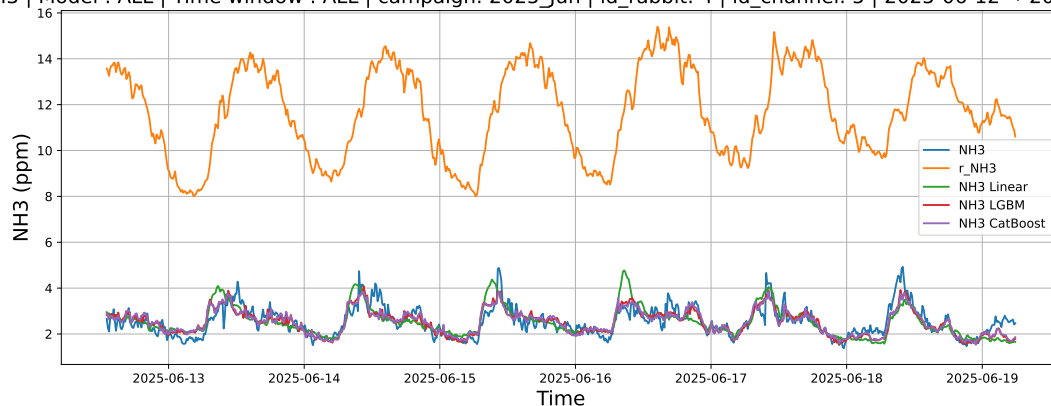
4.2.3.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



4.2.3.3 Time window : ALL

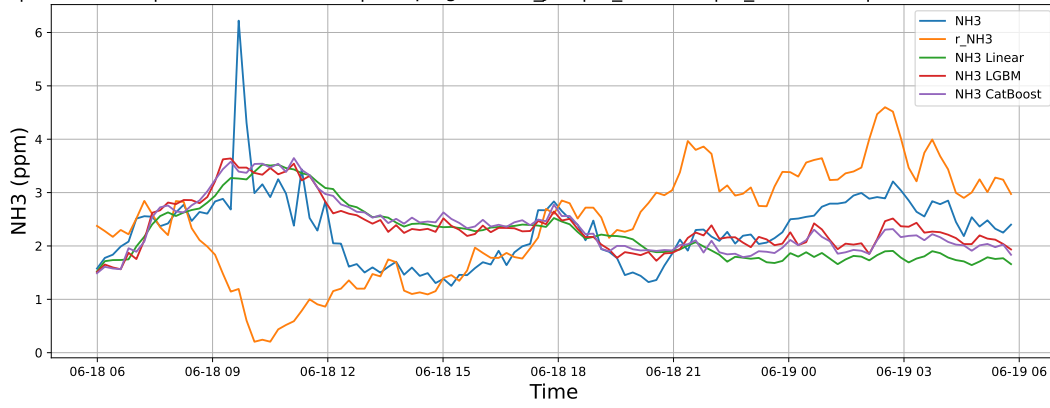
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



4.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

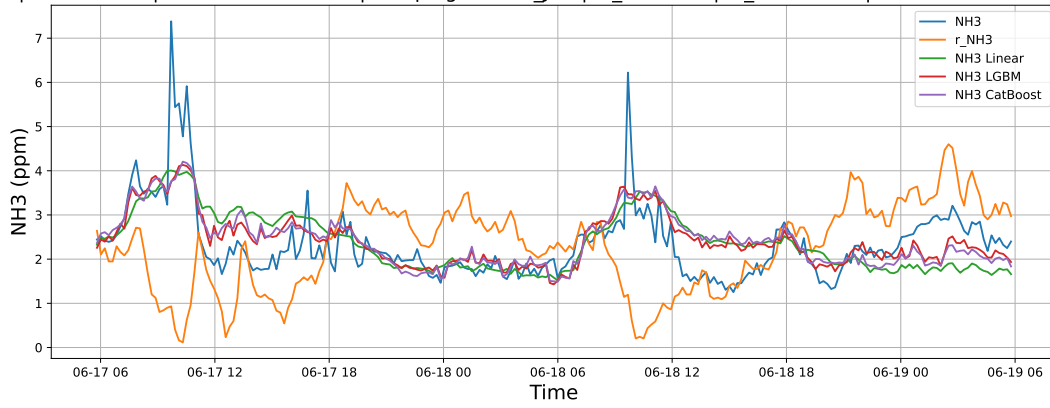
4.2.4.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



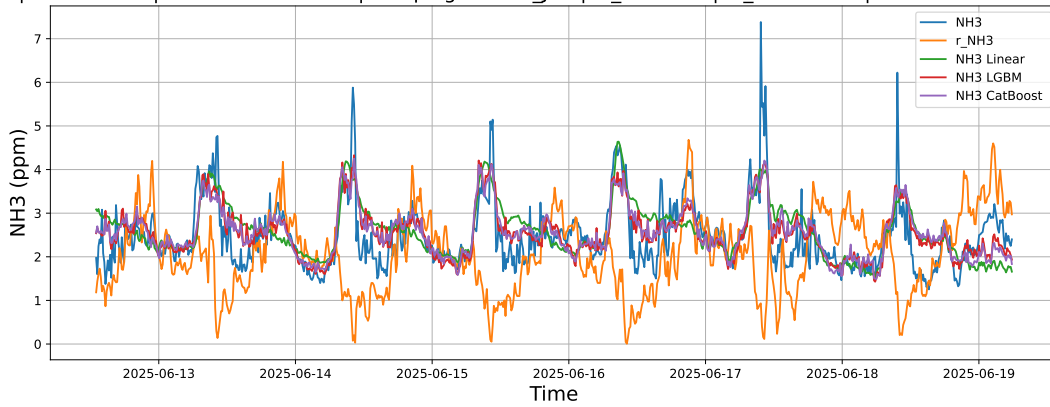
4.2.4.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



4.2.4.3 Time window : ALL

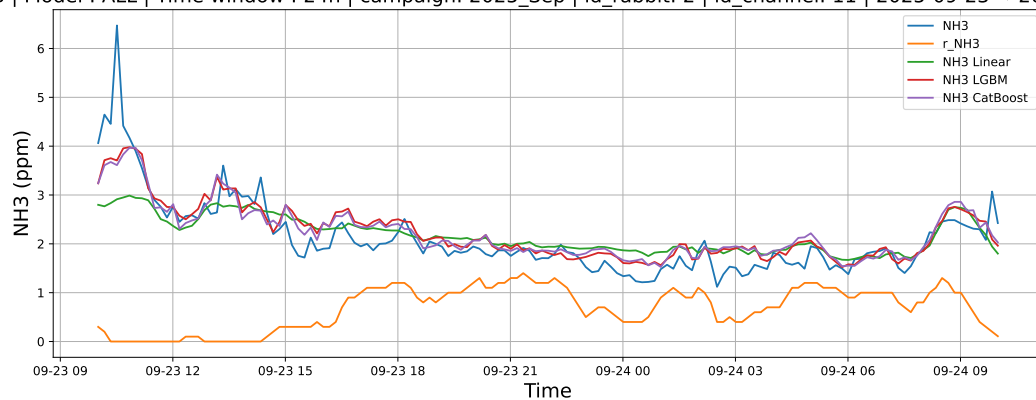
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



4.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

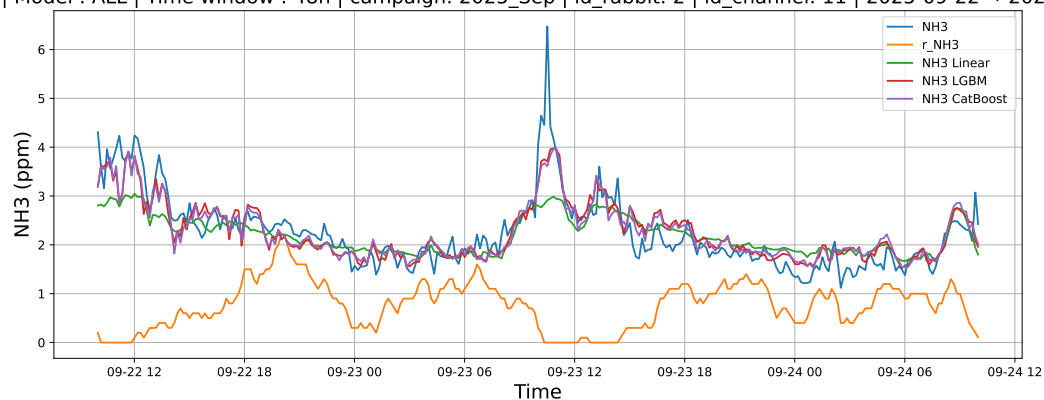
4.2.5.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



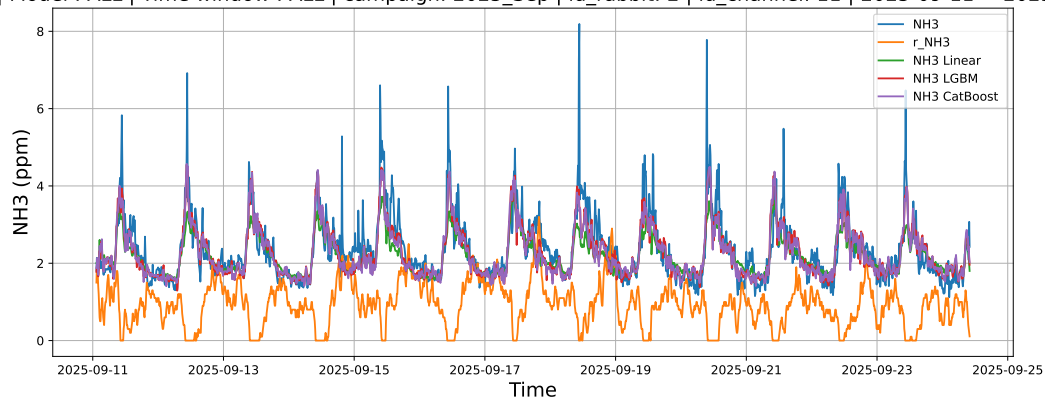
4.2.5.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



4.2.5.3 Time window : ALL

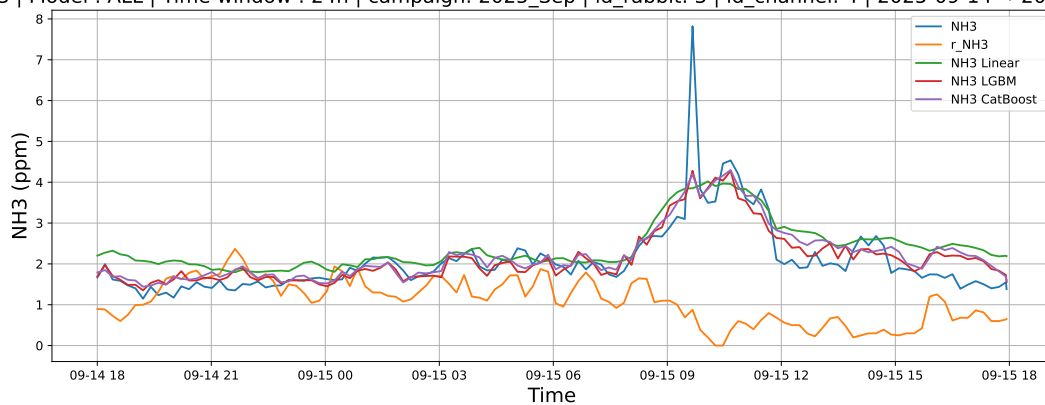
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



4.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

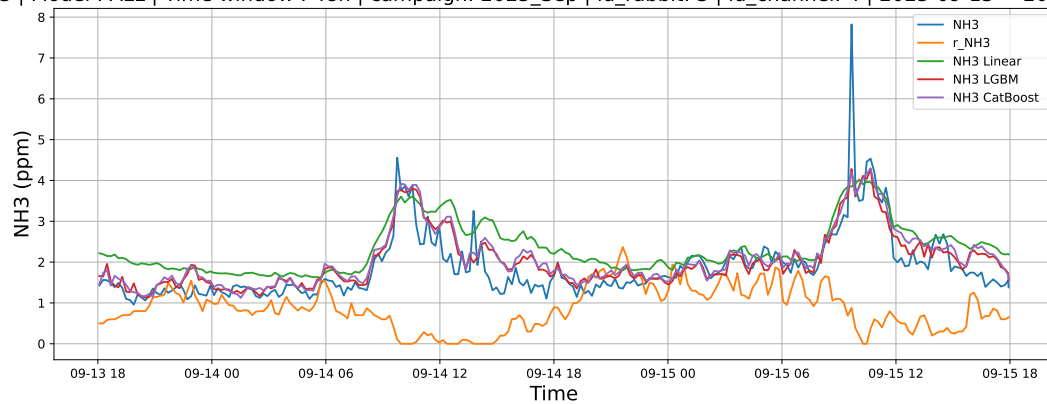
4.2.6.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



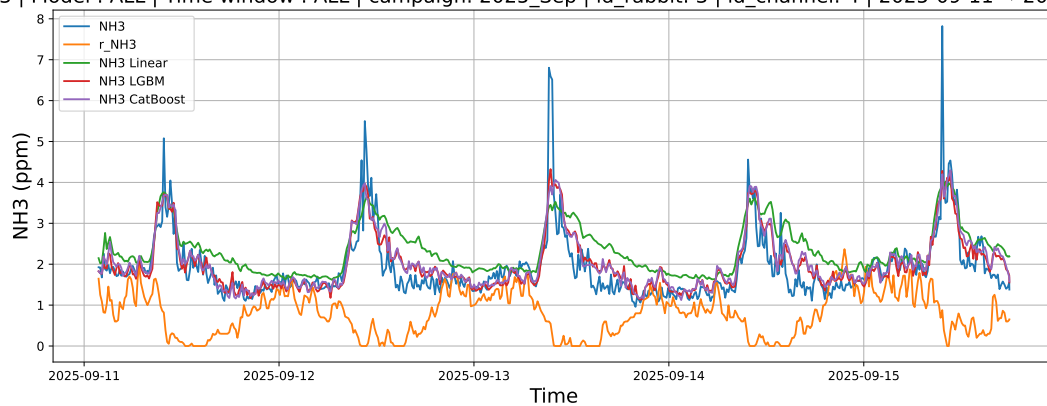
4.2.6.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



4.2.6.3 Time window : ALL

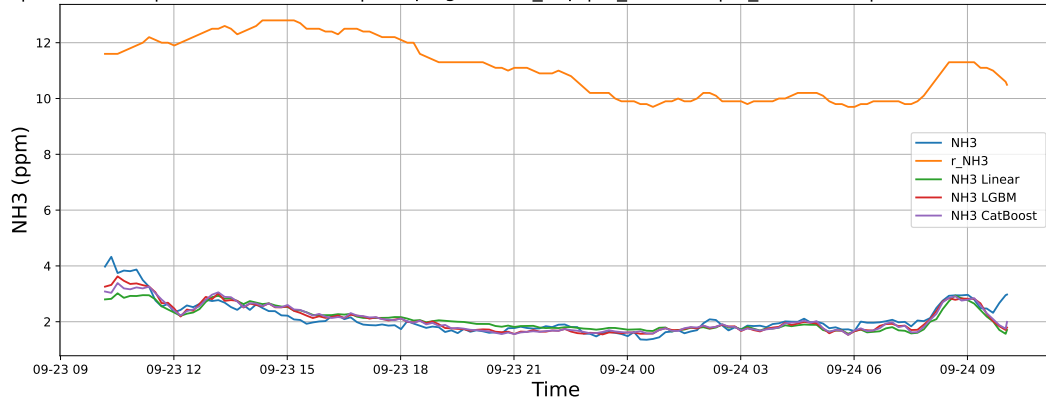
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



4.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

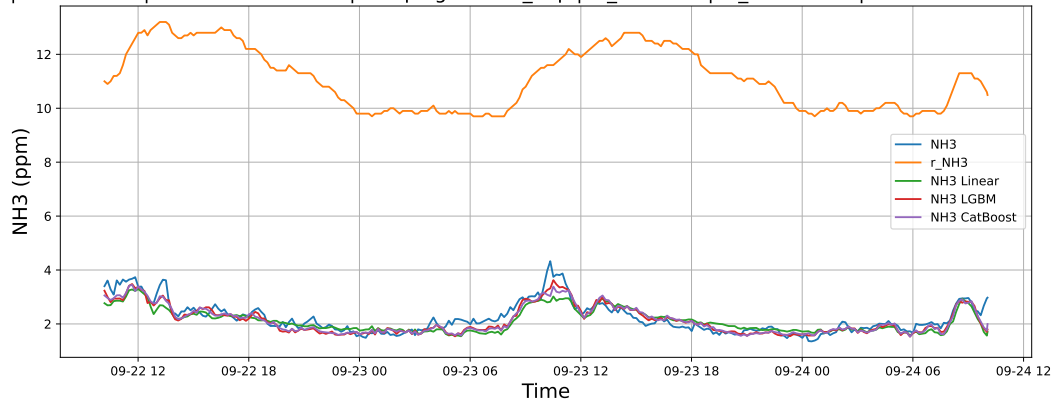
4.2.7.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



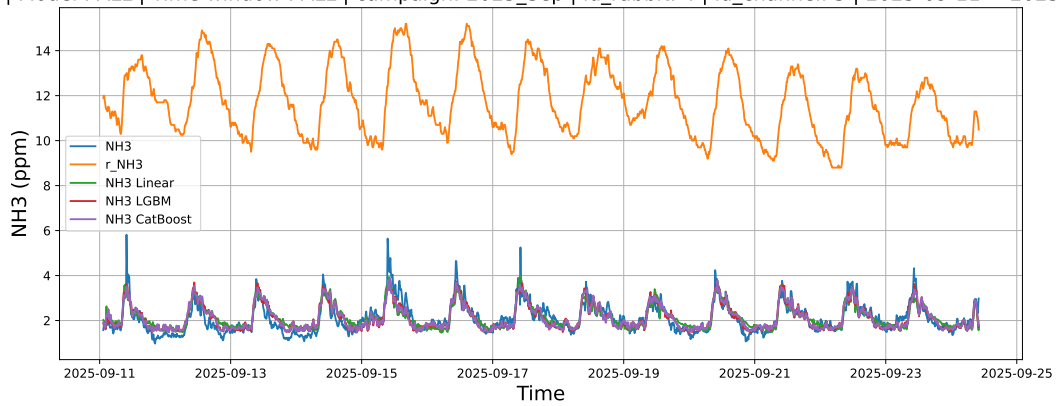
4.2.7.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



4.2.7.3 Time window : ALL

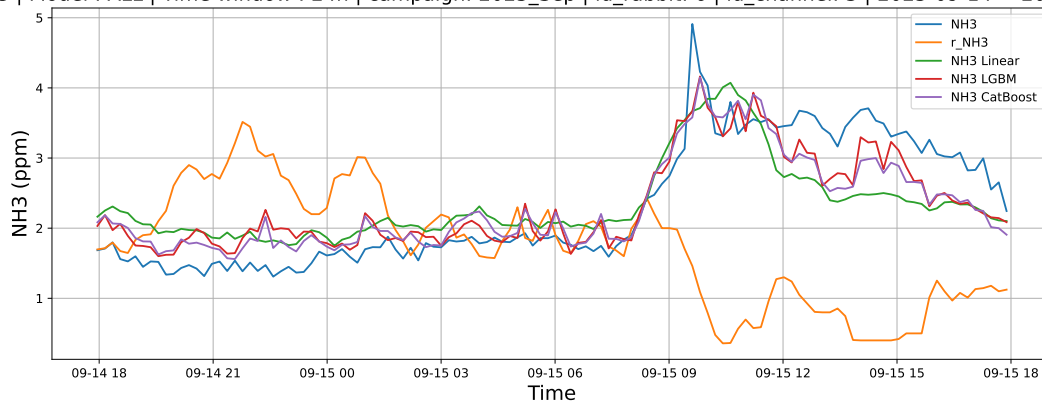
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



4.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

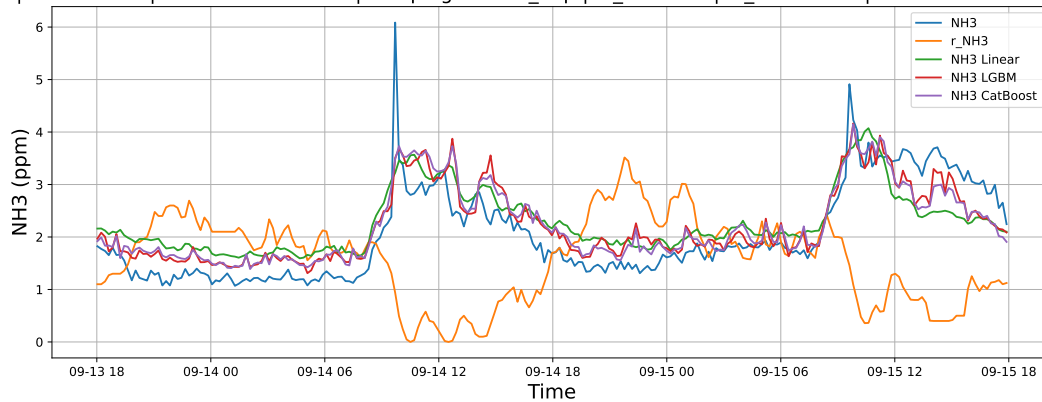
4.2.8.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



4.2.8.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



4.2.8.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

